

Uniform Inference for High-Frequency Data*

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Abstract

We develop uniform inference for conditional mean and quantile processes of high-frequency observations generated by a continuous-time state-space model. The functional estimators are formed as the collection of localized estimates over a growing number of shrinking time blocks. Their joint approximation requires balancing the accuracy of the local distributional approximation against variation in the latent state. A blockwise clipped martingale coupling exploits the chronological scalar structure to avoid the polynomial dimension penalty in a direct application of a general vector-valued Yurinskii bound. The resulting strong approximation accommodates rough states and heavy tailed observations. An application to SPY and Bitcoin data illustrates distinct patterns of return dispersion and trading activity around FOMC announcements.

Keywords: uniform inference, high-frequency data, martingale coupling.

JEL Codes: C14, C58, G12.

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1 Introduction

Financial markets adjust to new information through changes in both prices and other market indicators such as volumes and trading flows.^{1,2} Assessing these changes requires more than estimating a conditional moment at a selected instant: the timing, persistence, and uncertainty of the entire intraday pattern are often the objects of interest. High-frequency observations provide the local information needed for such assessments (see, e.g., [Jacod and Protter \(2012\)](#), [Aït-Sahalia and Jacod \(2014\)](#)). However, conventional pointwise confidence intervals do not control the probability of covering a path simultaneously, see [Jacod et al. \(2021\)](#) and [Bollerslev et al. \(2021\)](#). We develop a uniform inference framework that provides this simultaneous coverage for conditional means and quantiles of a broad class of noisy state observations.

Our setting builds on the general continuous-time state-space framework of [Li and Xiu \(2016\)](#), where the observation over a short interval is a transformation of the current latent state and a disturbance, up to an approximation error. The targets are distributional features of this transformation, not necessarily the state itself. This distinction accommodates, for example, the conditional mean of a volume observation and the conditional quantile of a return-based scale measure without specifying a parametric law of motion for the state. Identification of an economic state from these features remains model-specific, see applications in [Bollerslev et al. \(2018\)](#) and [Bugni et al. \(2023\)](#).

The statistical difficulty of performing uniform inference in such models comes from the interaction of two requirements. Each local block must contain enough observations for an accurate distributional approximation, while its time span must be short enough to track the state. The latter is particularly consequential for diffusion-driven states, whose paths need not be differentiable. Moreover, the number of blocks increases as the sampling interval shrinks. Gaussian limits at infinitely many times therefore do not provide a tight, fixed limiting process for the standardized estimation errors over the observation horizon, see [van der Vaart and Wellner \(1996\)](#). We instead approximate the entire collection by a sequence of conditionally Gaussian objects of increasing dimension.

Against this background, our main theoretical contribution is a blockwise Gaussian approximation tailored to this high-frequency block size tradeoff. By exploiting chronologically ordered scalar blocks, the construction avoids the polynomial dimension cost of a direct application of

¹Volumes and trading flows have observation mechanisms and normalizations different from those of prices. Consequently, inference methods developed for prices do not automatically provide inference for these indicators.

²Other market indicators deserve their own analysis. It is widely acknowledged that price and volumes may carry different aspects of information. For example, [Kandel and Pearson \(1995\)](#) argued that certain news may not significantly alter investors' average opinion, resulting in small price movement. However, it can lead to a substantial divergence of opinions among investors, prompting disagreements and subsequent trading activity between these disagreeing investors. Consequently, a surge in volume is observed but no significant change in price. Conversely, [Kyle \(1985\)](#) demonstrated that in the presence of asymmetric information, even a small trade can trigger a notable price change if the market impact is significant. In a nutshell, price reflects the average opinion, while trading activity also reflects the dispersion of opinion.

a general vector coupling bound. This permits block sizes that are compatible even with rough states, which has been a trending topic in the recent literature (see, e.g., [Gatheral et al. \(2018\)](#), [Fukasawa et al. \(2022\)](#), [Bolko et al. \(2023\)](#), [Wang et al. \(2023\)](#), and [Li et al. \(2025\)](#)). Quantile inference additionally permits observations without finite moments, subject to local density regularity, motivated by heavy-tailed returns in cryptocurrency markets, see [Kolokolov \(2022\)](#).

The relevant comparison is with a direct specialization of the vector-valued coupling inequality in [Li and Liao \(2020\)](#) and [Cattaneo et al. \(2025\)](#). For the same bounded, exact variance block array, the second order specialization incurs a polynomial cost in the number of blocks. Its zero-skewness refinement reduces, but does not remove, that cost. Section 2 makes the comparison explicit. Our result is thus a dimension-efficient approximation for a structured class of block statistics, rather than a general improvement on their martingale coupling. We build on their conditional coupling techniques and the smooth maximum bounds of [Chernozhukov et al. \(2014\)](#). The latter approximation of suprema, and the high-dimensional central limit theorem of [Chernozhukov et al. \(2017\)](#), address a weaker statistical objective and are important alternatives when only a maximum statistic is required. The distinction here is the local-sampling regime and its compatibility with a time-varying, potentially nonstationary state.

The quantile results provide a complementary route when moment-based inference is inappropriate. We establish a uniform Bahadur representation over all blocks for a fixed quantile level. The intrinsic empirical process remainder can be controlled almost surely adapting the simultaneous grid argument in [Bahadur \(1966\)](#) and [Wu \(2005\)](#). The full approximation remains in probability under our continuous-time specification. Because the leading scores are bounded, the fixed-quantile coupling requires no moments of the observations themselves. Under the stronger assumption of conditionally independent disturbances, the representation extends to a compact interval of quantile levels. A joint empirical process coupling then gives one density-scaled Brownian bridge for each block and retains the dependence across quantile levels. This process result justifies a feasible inversion procedure that evaluates empirical quantiles at nearby levels without estimating a density.

These results complement several strands of the high-frequency literature. Classical spot and integrated volatility analyses include [Foster and Nelson \(1996\)](#), [Comte and Renault \(1998\)](#), [Mykland and Zhang \(2009\)](#), and [Jacod and Rosenbaum \(2013\)](#). Closest in its uniform inferential objective, [Jacod et al. \(2021\)](#) develops volatility coupling for growing collections of local estimates. Our targets and observation formulation extend beyond volatility, while our block-wise martingale argument addresses the accompanying approximation problem. The fixed- k approach of [Bollerslev et al. \(2021\)](#) and the optimal range-based estimators of [Bollerslev et al. \(2024a\)](#) instead retain non-Gaussian local experiments for pointwise inference. Those papers study estimator optimality within local experiments, whereas our objective is simultaneous path coverage with growing local sample sizes. [Bollerslev et al. \(2018\)](#) and [Bugni et al. \(2023\)](#) provide a further connection between local distributional features and public news. For quantiles, [Shephard \(2022\)](#) studies infill medians of absolute returns under stochastic volatility. The

growing dimensional quantile process approximation in [Belloni et al. \(2019\)](#) is a close methodological antecedent. Our additional task is to control the joint effect of many shrinking time blocks and within-block state variation. The instantaneous conditional quantiles studied here are also distinct from quantiles of a volatility occupation distribution: the former describe the observation distribution at a given time, while the latter summarize the values taken by a path over time.

The numerical and empirical analyses show the usefulness of the results. In simulations, both studentized mean bands and quantile-level inversion yield coverage close to nominal level under various settings. The empirical illustration uses SPY and Bitcoin observations on eight FOMC announcement days in 2025. SPY exhibits a consistent second stage of adjustment during the press conference period and persistent trading near the market close. Bitcoin’s median-based return dispersion is more heterogeneous across dates. The application illustrates how uniform inference distinguishes the timing of changes in dispersion and trading activity.

The rest of the paper is organized as follows. Section 2 presents the model, the coupling results, and feasible confidence bands. Section 3 reports the Monte Carlo evidence. Section 4 gives the FOMC illustration, and Section 5 concludes. Proofs are collected in the Appendix.

Notation. We use $\|\cdot\|$ to denote the ℓ_2 -norm and use $\mathbb{1}\{\cdot\}$ to denote the indicator function. For two real numbers a and b , write $\min\{a, b\}$ as $a \wedge b$ and $\max\{a, b\}$ as $a \vee b$. For two positive sequences a_n and b_n , write $a_n \asymp b_n$ if $a_n/C \leq b_n \leq Ca_n$ for some finite constant $C \geq 1$.

2 Theory

2.1 Setting

We observe a data sequence $(Y_{n,i})_{1 \leq i \leq n}$, where $n \equiv \lfloor T/\Delta_n \rfloor$ within a fixed time span $[0, T]$. In what follows, we consider in-fill asymptotics, i.e., $\Delta_n \rightarrow 0$ and write $L_n \equiv \log(\Delta_n^{-1})$. Following the state-space formulation of [Li and Xiu \(2016\)](#), suppose that

$$Y_{n,i} = \mathcal{Y}(\zeta_{i\Delta_n}, \varepsilon_{n,i}) + R_{n,i}, \quad \text{for } 1 \leq i \leq n, \quad (1)$$

where $(\zeta_t)_{t \in [0, T]}$ is a càdlàg state process which takes values in an open set $\mathcal{Z}$ and is defined on some filtered probability space satisfying the usual conditions, denoted as $(\Omega^{(0)}, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P}^{(0)})$. The measurable transformation $\mathcal{Y}(\cdot, \cdot)$ represents a deterministic noisy transform of the *current* state $\zeta_{i\Delta_n}$ through a random disturbance $\varepsilon_{n,i}$ which takes values in a Polish space $\mathcal{D}$. The disturbance sequence is defined on $(\Omega^{(1)}, \mathcal{G}, \mathbb{P}^{(1)})$ and has common marginal law $\mathbb{P}_\varepsilon$. We work on the product space with $\Omega \equiv \Omega^{(0)} \times \Omega^{(1)}$, $\mathcal{H} \equiv \mathcal{F} \otimes \mathcal{G}$, and $\mathbb{P} = \mathbb{P}^{(0)} \otimes \mathbb{P}^{(1)}$. Thus the disturbance sequence is independent of the state, although its observations may be dependent across time. Conditioning on $\mathcal{F}$ fixes the entire state path. Regular conditional distributions are assumed to exist, and the conditioning sigma-fields used for coupling are countably generated modulo completion. These are the regularity conditions used in the conditional Strassen theorem of [Cattaneo et al. \(2025\)](#).

From an intuitive standpoint, the observation may depend on the state within a short interval rather than only at the current time. Given that state processes exhibit sufficient smoothness, information contained in the difference between past state and current state could be negligible as sampling interval shrinks. The remainder $R_{n,i}$ accommodates this difference, provided that it satisfies the uniform bound imposed below. It therefore permits the observation equation to be an approximation, hence renders model [1](#) more general than [Li and Xiu \(2016\)](#) and [Bollerslev et al. \(2018\)](#).

Three examples below clarify the economic targets and the scope of this model.

Example 1. *The simplest specification is when the transformation has an additive structure*

$$Y_{n,i} = \mu_{i\Delta_n} + \varsigma_{i\Delta_n} \varepsilon_{n,i} + R_{n,i}, \quad \text{for } 1 \leq i \leq n.$$

Here $\zeta_t = (\mu_t, \varsigma_t)$ and $\mathcal{Y}((\mu, \varsigma), \varepsilon) = \mu + \varsigma \varepsilon$. If the disturbance is centered with finite mean, the conditional mean of the leading observation is μ_t and coincides with its conditional median for a median-zero disturbance. Time-varying conditional means and scales arise naturally in models of trading volume and noisy observations of efficient prices, see [Liu and Tang \(2013\)](#), [Bollerslev et al. \(2018\)](#), and [Bugni et al. \(2023\)](#). The observation may also be a preliminary local statistic rather than a raw price or volume. For example, logarithms of positive scale statistics in a fixed- k experiment can have an additive leading representation, where the approximation errors are uniformly negligible, see [Bollerslev et al. \(2021\)](#).

Example 2. *The proposed state-space model can be applied to characterize a wide range of price dynamics. Motivated by [Kolokolov \(2022\)](#), consider the log price driven by a symmetric stable Lévy process*

$$P_t = \int_0^t \mu_s ds + \int_0^t \varsigma_s dL_s, \quad \text{for } t \in [0, T],$$

where μ is the drift, ς is the stochastic volatility, L is a stable process with Blumenthal–Gettoor index $\beta \in (0, 2]$. For $\beta = 2$, take L to be standard Brownian motion. For $0 < \beta < 2$, take L to be a symmetric stable Lévy process normalized by $\mathbb{E}[\exp\{iu(L_{t+s} - L_t)\}] = \exp\{-s|u|^\beta\}$. This normalization fixes the arbitrary allocation of scale between ς and L . In particular, $\beta = 1$ gives the standard Cauchy process. The normalized squared return $Y_{n,i} = \Delta_n^{-2/\beta} (P_{(i+1)\Delta_n} - P_{i\Delta_n})^2$ over each observation window $(i\Delta_n, (i+1)\Delta_n]$ can be written as

$$Y_{n,i} = \Delta_n^{-2/\beta} \left(\int_{i\Delta_n}^{(i+1)\Delta_n} \mu_s ds + \int_{i\Delta_n}^{(i+1)\Delta_n} \varsigma_s dL_s \right)^2.$$

The scaled Lévy increments $\varepsilon_{n,i} \equiv \Delta_n^{-1/\beta} (L_{(i+1)\Delta_n} - L_{i\Delta_n})$ are i.i.d. with a non-degenerate distribution. Therefore, upon expanding above display and collecting dominant terms, the normalized squared return can be rewritten in model [\(1\)](#) by setting

$$\zeta_t = \varsigma_t, \quad \varepsilon_{n,i} = \Delta_n^{-1/\beta} (L_{(i+1)\Delta_n} - L_{i\Delta_n}), \quad \mathcal{Y}(\varsigma, \varepsilon) = (\varsigma \varepsilon)^2.$$

Its mean equals ζ_t^2 in the Brownian case. For $\beta < 2$, its mean is infinite, but its conditional quantiles remain finite. For example, its median is $\zeta_t^2(F_\beta^{-1}(3/4))^2$, which could serve as a scale feature. The remainder contains

$$R_{n,i} = \Delta_n^{-2/\beta} \left(\int_{i\Delta_n}^{(i+1)\Delta_n} \mu_s ds + \int_{i\Delta_n}^{(i+1)\Delta_n} (\zeta_s - \zeta_{i\Delta_n}) dL_s \right)^2 \\ + 2\Delta_n^{-2/\beta} \left(\int_{i\Delta_n}^{(i+1)\Delta_n} \mu_s ds + \int_{i\Delta_n}^{(i+1)\Delta_n} (\zeta_s - \zeta_{i\Delta_n}) dL_s \right) \times \zeta_{i\Delta_n} (L_{(i+1)\Delta_n} - L_{i\Delta_n}).$$

For the Brownian case, [Jacod et al. \(2021\)](#) and [Bollerslev et al. \(2021\)](#) show that, under fairly weak regularity condition, $\max_{1 \leq i \leq n} |R_{n,i}| = O_p(\Delta_n^r)$ for some $r > 0$. For related spot-volatility approximations under $\beta < 2$, see [Yan et al. \(2026\)](#).

Example 3. Let $\Delta_i^n N = N_{(i+1)\Delta_n} - N_{i\Delta_n}$ denote number of transactions during each observation window $(i\Delta_n, (i+1)\Delta_n]$. Following [Christensen and Kolokolov \(2024\)](#), it is cogent to model trading flows as a Cox process, see [Cox \(1955\)](#). The sparseness property (see, e.g., Section 5.4.1 in [Ross \(1995\)](#)) yields (i) $\mathbb{P}(\Delta_i^n N \geq 2 \mid \mu) = o(\Delta_n)$ and (ii) $\mathbb{P}(\Delta_i^n N = 1 \mid \mu) = \Delta_n \mu_{i\Delta_n} + o(\Delta_n)$. This naturally suggests an approximation by a mixed Bernoulli random variable with parameter $\Delta_n \mu_{i\Delta_n}$.³ We maintain the “heavy traffic” assumption in [Kingman \(1961\)](#) and [Hayashi et al. \(2011\)](#) that $\Delta_n \mu_t = O_p(1)$. Above display shows $\Delta_i^n N$ can be expressed in model (1) by setting

$$\zeta_t = \Delta_n \mu_t, \quad \mathcal{V}(\zeta, \varepsilon) = \mathbb{1}\{\varepsilon < \zeta\}, \quad \varepsilon_{n,i} \sim \text{Uniform}(0, 1),$$

where $R_{n,i} = \Delta_i^n N - \mathbb{1}\{\varepsilon_{n,i} < \Delta_n \mu_{i\Delta_n}\}$. Note that in the Trade and Quote (TAQ), each trade is recorded with a precision of nanoseconds (10^{-9} seconds) since September 18, 2017. Consequently, the mixed Bernoulli approximation here exactly matches empirical data: a binary sequence is observed indicating whether a trade has transpired within each preceding nanosecond window.⁴ Therefore, we shall assume additionally the trading flow process to have refractoriness (see, e.g., [Citi et al. \(2014\)](#)), which gives $\max_{1 \leq i \leq n} |R_{n,i}| = O_p(\Delta_n)$.

As shown in above examples, the residuals are usually uniformly negligible as $\Delta_n \rightarrow 0$ under various model specifications. We impose the following envelope for residuals and restrict its rate in theorems.

Assumption 1. *There exists a deterministic sequence $B_{n,R}$ such that $\max_{1 \leq i \leq n} |R_{n,i}| = O_p(B_{n,R})$.*

Following [Li and Xiu \(2016\)](#) and [Bollerslev et al. \(2018\)](#), although our primary interest lies in unobservable states, we estimate distributional features of the transformed observations

³The approximation has been explored from a different direction as well, see, e.g., Section 1.6 of [Karr \(1991\)](#) where they discuss the optimal approximation of a Bernoulli point process by a Poisson process.

⁴Trading flows allow to be analyzed at a much higher frequency and reflects the speed at which market participants react to and incorporate new information into their idiosyncratic trading strategies. [Du and Zhu \(2017\)](#) shows a surge in trading intensity usually indicates higher level of information flow and potentially reflects real-time changes in market sentiment or news announcements that influence trading activity.

rather than recovering the state itself. The identification of the state from these distribution features should be model-specified. In particular, the targets are the conditional mean and quantile processes: for $t \in [0, T]$ and $\tau \in (0, 1)$,

$$G_t \equiv \int_{\mathcal{D}} \mathcal{Y}(\zeta_t, \varepsilon) \mathbb{P}_\varepsilon(d\varepsilon), \quad Q_t(\tau) \equiv \inf\{x \in \mathbb{R} : \mathbb{P}_\varepsilon(\mathcal{Y}(\zeta_t, \varepsilon) \leq x) \geq \tau\}.$$

The mean is considered only when it exists. To construct functional estimators for these processes, we partition the sample into m_n chronologically ordered, nonoverlapping blocks by dividing the whole index set $\{1, \dots, n\} = \bigcup_{j=1}^{m_n} \mathcal{I}_{n,j}$, where $\mathcal{I}_{n,j}$ denotes the set of $k_{n,j}$ consecutive indices contained in the j th block. Specifically, we define $\iota(i, j) \equiv \min \mathcal{I}_{n,j} + i - 1$ as the i th index in the j th block. Consequently, we have $\mathcal{I}_{n,j} = \{\iota(i, j) : 1 \leq i \leq k_{n,j}\}$, which spans time interval $\mathcal{T}_{n,j} \equiv [\iota(1, j)\Delta_n, \iota(1, j+1)\Delta_n]$. We set $\mathcal{T}_{n,1} \equiv [0, \iota(k_{n,1}, 1)\Delta_n]$ and $\mathcal{T}_{n,m_n} \equiv [\iota(1, m_n)\Delta_n, T]$ for first and the last block so that the block intervals partition $[0, T]$.

2.2 Uniform inference for conditional mean process

Given that G_t is simply the conditional mean of transformed state ζ_t , it naturally suggests a functional estimator by collecting local average within each block, see also [Jacod et al. \(2021\)](#). Following the same idea, set

$$\hat{G}_{n,t} \equiv \hat{G}_{n,j}, \quad \text{for } t \in \mathcal{T}_{n,j} \text{ and } 1 \leq j \leq m_n, \text{ where } \hat{G}_{n,j} \equiv \frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} Y_{n,i}.$$

For ease of presentation, we write $g(z) \equiv \int_{\mathcal{D}} \mathcal{Y}(z, \varepsilon) \mathbb{P}_\varepsilon(d\varepsilon)$ and $\sigma(z)^2 \equiv \int_{\mathcal{D}} \mathcal{Y}(z, \varepsilon)^2 \mathbb{P}_\varepsilon(d\varepsilon) - (\int_{\mathcal{D}} \mathcal{Y}(z, \varepsilon) \mathbb{P}_\varepsilon(d\varepsilon))^2$, so $G_t = g(\zeta_t)$ and $\sigma_t^2 = \sigma(\zeta_t)^2$. Under the pointwise conditions of [Bollerslev et al. \(2018\)](#), an appropriately undersmoothed local estimator satisfies

$$\sqrt{k_{n,j}}(\hat{G}_{n,t} - G_t) \xrightarrow{\mathcal{L}\text{-s}} \mathcal{MN}(0, \sigma_t^2), \quad (2)$$

Here t is fixed and j denotes its containing block, $\xrightarrow{\mathcal{L}\text{-s}}$ denotes stable convergence, and $\mathcal{MN}$ denotes mixed Gaussian distribution. Besides $k_{n,j} \rightarrow \infty$ and $k_{n,j}\Delta_n \rightarrow 0$, centering at G_t requires the standardized local bias and observation remainder to be negligible. Under H -regularity in Assumption 2, a typical bias restriction is $\sqrt{k_{n,j}}(k_{n,j}\Delta_n)^H \rightarrow 0$.

Above results can be easily extended to the joint convergence of $\hat{G}_{n,(\cdot)}$ on a *finite* set of time points. However, difficulty arises in extending this to estimation of the entire process on a *continuum* set of indices, which is primarily due to the absence of functional central limit theorems. Pointwise central limit theorem (2) and non-overlapping blocks further imply that process of estimation error have a path structure similar to a Gaussian white noise, hence is not asymptotically equicontinuous in probability on $[0, T]$ (see Section 1.5 in [van der Vaart and Wellner \(1996\)](#)). The uniform inference problem based on this type of functional estimators is non-Donsker in nature. That being said, such non-Donsker problems that commonly arise from uniform inference in nonparametric settings, can be addressed using strong approximation of

the functional estimators by variables with known finite-sample distributions, see, e.g., [Chernozhukov et al. \(2017\)](#) for the independent data, [Li and Liao \(2020\)](#) and [Cattaneo et al. \(2025\)](#) for generalization to time series data. Our objective is a sup-norm coupling of the entire standardized error, with a conditionally Gaussian vector whose dimension grows with m_n . This is stronger than approximating only the distribution of its maximum ([Chernozhukov et al. \(2013\)](#) and [Chernozhukov et al. \(2014\)](#)). The following conditions specify the state regularity, the tail envelope, and the predictable variance approximation used for that purpose.

Assumption 2. *There exist a sequence $(T_m)_{m \geq 1}$ of stopping times increasing to infinity, a sequence of compact subsets $(\mathcal{K}_m)_{m \geq 1}$ of $\mathcal{Z}$, and a sequence $(K_m)_{m \geq 1}$ of positive constants, and a positive constant $H \in (0, 1]$, such that for each $m \geq 1$ it holds:*

(i) $\zeta_{t \wedge T_m}$ takes values in $\mathcal{K}_m$; for all $s < t$ with $s, t \in \mathcal{T}_{n,j}$ where $1 \leq j \leq m_n$, and for each $p \geq 1$, there exists some constant $K_{m,p}$ such that

$$\mathbb{E} \left[\sup_{u, u' \in [s, t]} \|\zeta_{u \wedge T_m} - \zeta_{u' \wedge T_m}\|^p \right] \leq K_{m,p} |t - s|^{pH};$$

(ii) for $z, z' \in \mathcal{K}_m$, $\sigma(z)^{-2} \vee \sigma(z')^{-2} \leq K_m$ and $\mathbb{E}[|\mathcal{Y}(z, \varepsilon) - \mathcal{Y}(z', \varepsilon)|^2] \leq K_m \|z - z'\|^2$.

Assumption 2 imposes some regularity conditions on the state process and the transformation of random disturbance, which allow for state processes whose dynamics need not be stationary or parametrically specified. Specifically, condition (i) requires state process to be locally taken value in compact set. Condition (i) also imposes the smoothness of state process *within* each block. Namely, it requires paths of the state process to be H -Hölder continuous under the L_p -norm. Note that H is allowed to be smaller than $1/2$, which means we do not rule out rougher states, as motivated by [Gatheral et al. \(2018\)](#) among others. Condition (ii) requires the variance to be bounded away from zero and makes both $g(\cdot)$ and $\sigma(\cdot)$ Lipschitz in the state, so the target mean and standard deviation inherit the same regularity exponent H . No additional smoothness is imposed. Such condition can be verified for examples in [Section 2.1](#).

Assumption 3. *For $1 \leq i \leq n$, let $U_{n,i} \equiv \mathcal{Y}(\zeta_{i\Delta_n}, \varepsilon_{n,i}) - G_{i\Delta_n}$ be the centered disturbances and $\mathcal{H}_{n,i} \equiv \mathcal{F} \vee \sigma(\varepsilon_{n,k} : k \leq i)$. There exist deterministic sequences $B_{n,1}, B_{n,2}, d_n$ satisfying*

- (i) $\mathbb{E}[U_{n,i} | \mathcal{H}_{n,i-1}] = 0$ and $\mathbb{E}[\max_{1 \leq i \leq n} |U_{n,i}|^{2+q}] \leq K_q B_{n,q}$ for $q = 1, 2$;
- (ii) it holds almost surely that $\max_{1 \leq i \leq n} \mathbb{E}[U_{n,i}^2 | \mathcal{H}_{n,i-1}] \leq K$ and

$$\max_{1 \leq j \leq m_n} \left| \frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} \{\mathbb{E}[U_{n,i}^2 | \mathcal{H}_{n,i-1}] - \sigma_{i\Delta_n}^2\} \right| \leq K d_n.$$

Condition (i) implies after centering, the transformed residuals $(U_{n,i})_{1 \leq i \leq n}$ is a martingale difference relative to $(\mathcal{H}_{n,i})_{1 \leq i \leq n}$, and further imposes envelopes for the unconditional third and fourth moments. The envelopes may increase with Δ_n^{-1} , and the theorem restricts their growth. Condition (ii) is a separate requirement on conditional second moments and blockwise

predictable variance. We highlight that under Assumption 3, the uniform approximation problem is essentially close to coupling all blockwise sums of martingale difference. For bounded residuals, the maximal moment envelopes can be constant. For sub-Gaussian residuals, a union bound and tail integration (see Lemma 2.2.2 in [van der Vaart and Wellner \(1996\)](#)) give $B_{n,1} \asymp L_n^{3/2}$ and $B_{n,2} \asymp L_n^2$. If the disturbances are also independent across observations, conditional on the state, predictable variance matching is exact and (ii) holds with $d_n = 0$.

Theorem 1. *Suppose Assumptions 1-3 hold, and $k_{n,j} \asymp \Delta_n^{-\varrho}$ uniformly for $1 \leq j \leq m_n$ such that $0 < \varrho < \frac{2H}{1+2H}$. Assume further $(B_{n,1}^{2/3} \wedge B_{n,2}^{1/2})\Delta_n^{\varrho/3} \vee d_n = o(L_n^{-5})$ and $B_{n,R} = o(\Delta_n^{\varrho/2}L_n^{-1})$. Then there exists a mixed Gaussian vector $Z_n = (Z_{n,j})_{1 \leq j \leq m_n} \in \mathbb{R}^{m_n}$ on an extension space with*

$$Z_n \mid \mathcal{F} \sim \mathcal{N}(0, \text{diag}\{\sigma_{\iota(1,j)\Delta_n}^2 : 1 \leq j \leq m_n\}),$$

such that

$$\max_{1 \leq j \leq m_n} \sup_{t \in \mathcal{I}_{n,j}} |\sqrt{k_{n,j}}(\hat{G}_{n,t} - G_t) - Z_{n,j}| = o_p(L_n^{-1}).$$

The theorem states a joint Yurinskii's coupling for the entire functional estimator. The maximum norm error $o_p(L_n^{-1})$ is a sufficient margin for inference, not an optimal approximation rate. For bounded or sub-Gaussian residuals with exact predictable variance, the moment restrictions are automatic under such polynomial block growth.

The construction of such coupling in nonparametric inference literature is usually through a “stacking and padding” transformation. For a transparent comparison, consider equal size blocks and bounded residuals with exact predictable-variance matching. Denote temporarily that

$$X_{n,i} \equiv U_{n,i} \sum_{j=1}^{m_n} \frac{e_{n,j}}{\sqrt{k_{n,j}}} \mathbb{1}\{i \in \mathcal{I}_{n,j}\}, \quad \text{for } 1 \leq i \leq n,$$

where $e_{n,j} \in \mathbb{R}^{m_n}$ is the j th coordinate vector. Then $\mathbb{E}[X_{n,i} \mid \mathcal{H}_{n,i-1}] = 0$, so the construction preserves the martingale difference property, and $\sum_{i=1}^n X_{n,i}$ collects the normalized scalar blockwise sums. The target is then a Yurinskii's coupling for this growing dimensional martingale, which has been studied in [Li and Liao \(2020\)](#) and [Cattaneo et al. \(2025\)](#). The aggregate third-order moment quantity in Theorem 2.1 of [Cattaneo et al. \(2025\)](#) is of order $m_n/\sqrt{k_{n,j}}$, and the analogous fourth-order quantity is of order $m_n/k_{n,j}$. Directly applying the respective branches of their bound at precision $o_p(L_n^{-1})$ therefore gives the sufficient restrictions

$$\begin{cases} m_n L_n^4 / \sqrt{k_{n,j}} \rightarrow 0, & \text{for general second order approximation;} \\ m_n L_n^{11/2} / k_{n,j} \rightarrow 0, & \text{with zero conditional third moments.} \end{cases}$$

The moment orders follow because each $X_{n,i}$ has one nonzero coordinate. Since $m_n \asymp \Delta_n^{-(1-\varrho)}$, the polynomial lower bounds on ϱ are respectively $2/3$ and $1/2$. Combining them with the block-averaging state restriction $\varrho < 2H/(1+2H)$ gives Table 1. These are sufficient regimes delivered by different approaches.

Table 1: Polynomial regimes in a matched block-coupling benchmark

Coupling construction	Feasible block-size range	State smoothness
Direct CMU: second order	$2/3 < \varrho < 2H/(1 + 2H)$	$H > 1$
Direct CMU: third order	$1/2 < \varrho < 2H/(1 + 2H)$	$H > 1/2$
Blockwise construction, Lemma 1	$0 < \varrho < 2H/(1 + 2H)$	$H > 0$

Note: CMU refers to Theorem 2.1 of [Cattaneo et al. \(2025\)](#). Its third-order branch additionally requires zero conditional third moments here. Logarithmic restrictions are suppressed. The first row has no feasible range under the maintained $H \leq 1$. The table compares direct applications to the displayed stacking with the chronological construction used in this paper.

It is worth noting that direct vector martingale coupling methods in Table 1 do not deliver a feasible block-size range at $H = 1/2$, the case of continuous Itô semimartingale benchmark. Our construction avoids this polynomial cost by retaining the scalar conditional block law. This is a meaningful improvement for the intended experiment: the block size can be small enough to track a rough state while still supporting simultaneous Gaussian approximation. We do not claim our results as an improvement of existing rate in those papers, it should be posed as another strategy targeting this class of blockwise sum approximation problems.⁵

There are also alternative approximation tools. Under the special case of conditional independence, grouping observations by within-block position maintains independence structure,

$$X_{n,i}^* \equiv \sum_{j=1}^{m_n} U_{n,\iota(i,j)} \frac{e_{n,j}}{\sqrt{k_{n,j}}} = \left(\frac{U_{n,\iota(i,1)}}{\sqrt{k_{n,1}}}, \frac{U_{n,\iota(i,2)}}{\sqrt{k_{n,2}}}, \dots, \frac{U_{n,\iota(i,m_n)}}{\sqrt{k_{n,m_n}}} \right)^\top,$$

permits an application of Corollary 4.1 in [Chernozhukov et al. \(2014\)](#) and Proposition 2.1 in [Chernozhukov et al. \(2017\)](#), with effective sample size of order $k_{n,j}$, see the proof of Theorem 3 in [Jacod et al. \(2021\)](#). Under bounded residuals its sufficient maximum-distribution bound has order $(L_n^7/k_{n,j})^{1/6}$, which allows for the case when $H = 1/2$. However, this bound approximates the distribution of the maximum rather than coupling the entire vector. Its independent-array construction also cannot be transferred unchanged to dependent chronological blocks. Scalar exponential-tail couplings, including [Mason and Zhou \(2012\)](#) and [Fan et al. \(2020\)](#), provide another route and can yield sharper rates under their exponential tail conditions.

The next lemma states the conditional neighborhood comparison used in our construction. Its proof combines a diagonal smooth test adaptation of [Cattaneo et al. \(2025\)](#)’s martingale argument with smooth-maximum derivatives in [Chernozhukov et al. \(2014\)](#). It allows the coupling failure probability to be made exponentially small without putting a polynomial number-of-blocks factor into the displacement.

⁵We acknowledge that [Cattaneo et al. \(2025\)](#) requires $d_n = o(L_n^{-3})$ while we require $d_n = o(L_n^{-5})$. Therefore, our variance stabilization condition in Assumption 3(ii) is stronger by logarithmic factors. This difference disappears when $d_n = 0$, and is immaterial for polynomially vanishing d_n .

Lemma 1. Let $(U_i)_{1 \leq i \leq k}$ be scalar martingale differences with respect to filtration $(\mathcal{A}_i)_{0 \leq i \leq k}$. For some $b \geq 1$ and $d \geq 0$, suppose $|U_i| \leq Cb/\sqrt{k}$, $\mathbb{E}[U_i^2 | \mathcal{A}_{i-1}] \leq C/k$, and $|\sum_{i=1}^k \mathbb{E}[U_i^2 | \mathcal{A}_{i-1}] - 1| \leq Cd$ almost surely. Let $W \equiv \sum_{i=1}^k U_i$. There is a constant K_C such that for every fixed $\ell \geq 1$, with

$$\eta \equiv K_C \left\{ \left(\frac{(b^3 + \ell^{3/2})\ell^{7/2}}{\sqrt{k}} \right)^{1/4} + (d\ell^2)^{1/3} \right\},$$

then simultaneously for all $B \in \mathcal{B}(\mathbb{R})$ it holds

$$\mathbb{P}(W \in B | \mathcal{A}_0) - \nu(B^\eta | \mathcal{A}_0) \leq 2e^{-\ell},$$

where $B^\eta \equiv \{x : \inf_{y \in B} |x - y| \leq \eta\}$ denotes the η -enlargement for Borel B and $\nu(\cdot | \mathcal{A}_0) = \int_{(\cdot)} \phi(x) dx$ is $\mathcal{A}_0$ -conditionally standard Gaussian measure.

Conditional Strassen theorem (Lemma SA.1 in [Cattaneo et al., 2025](#)) turns above neighborhood inequality into a scalar Gaussian coupling. The variable can be selected using the block, its initial history, and fresh auxiliary uniforms. In the theorem proof, stopped clipping and variance repair first produce bounded martingale differences. Applying Lemma 1 successively, conditional on the state and the preceding enlarged history, yields Gaussian coordinates that are independent given $\mathcal{F}$ even when the original blocks are dependent. The common precision is fixed before selection, so a sequence of couplings is obtained rather than a different coupling for every probability threshold. For bounded residuals, Lemma 1 gives a sufficient approximation error of order $(k_{n,j})^{-1/8} L_n^{5/4} + (d_n L_n^2)^{1/3}$ with dependence on the growing block count being logarithmic through the chosen precision.

The covariance of Z_n depends on unknown state quantities, so the coupling is not itself a nuisance-free finite-sample distribution. Studentization in Corollary 1 provides a feasible reference. The proof also establishes the uniform accuracy of the variance estimator required for studentization.

Corollary 1. Under assumptions of Theorem 1, let $\text{cv}_{n,\alpha}$ be the $(1-\alpha)$ -quantile of the maximum of an $\mathbb{R}^{m_n}$ -dimensional folded standard Gaussian variable, and define

$$\hat{\sigma}_{n,j}^2 \equiv \Delta_n \vee \left(\frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} (Y_{n,i} - \hat{G}_{n,j})^2 \right).$$

Then it holds

$$\mathbb{P} \left(G_t \in \left[\hat{G}_{n,t} \pm \text{cv}_{n,\alpha} \frac{\hat{\sigma}_{n,j}}{\sqrt{k_{n,j}}} \right] \text{ for all } t \in \mathcal{T}_{n,j}, 1 \leq j \leq m_n \right) = 1 - \alpha + o(1).$$

The critical value can be computed analytically: $\text{cv}_{n,\alpha} = \Phi^{-1}((1 + (1 - \alpha)^{1/m_n})/2)$. It differs from a pointwise normal critical value by the multiplicity adjustment across blocks and is of order $\sqrt{\log(2m_n)}$. The conclusion is asymptotic simultaneous coverage and the truncation Δ_n in the variance estimate prevents division by zero asymptotically.

2.3 Uniform inference for conditional quantile process

Quantiles describe distributional features even when the moments needed for mean inference fail to exist.⁶ Their regular Gaussian approximation nevertheless requires a positive, sufficiently regular density locally at the quantile of interest, see [Koenker and Bassett Jr \(1978\)](#) and [Koenker \(2005\)](#). Following [Shephard \(2022\)](#), for $1 \leq j \leq m_n$, we define the blockwise infill quantile as

$$\widehat{Q}_{n,j}(\tau) = \arg \min_{q \in \mathbb{R}} \sum_{i \in \mathcal{I}_{n,j}} \rho_\tau(Y_{n,i} - q), \quad \rho_\tau(u) \equiv u(\tau - \mathbb{1}\{u < 0\}),$$

where we use the smallest one if the minimizer is not unique. The sample objective is then well defined for finite observations without an integrability assumption on the check loss. Similar as for the conditional mean, the functional estimator of the whole process is constructed as

$$\widehat{Q}_{n,t}(\tau) \equiv \widehat{Q}_{n,j}(\tau), \quad \text{for } t \in \mathcal{T}_{n,j} \text{ and } 1 \leq j \leq m_n.$$

We use a Bahadur representation to linearize these estimators, see [Bahadur \(1966\)](#) and [Ghosh \(1971\)](#) for independent data, [Hesse \(1990\)](#) and [Wu \(2005\)](#) for weakly dependent data. Our result is a uniform Bahadur representation that holds simultaneously over all blocks, which can be further extended to a compact interval of quantile levels.

Assumption 4. *There exist a sequence $(T_m)_{m \geq 1}$ of stopping times increasing to infinity, a sequence of compact subsets $(\mathcal{K}_m)_{m \geq 1}$ of $\mathcal{Z}$, and a sequence $(K_m)_{m \geq 1}$ of positive constants, and a positive constant $H \in (0, 1]$, such that for each $m \geq 1$ it holds:*

(i) $\zeta_{t \wedge T_m}$ takes values in $\mathcal{K}_m$; for all $s < t$ with $s, t \in \mathcal{T}_{n,j}$ where $1 \leq j \leq m_n$, and for each $p \geq 1$, there exists some constant $K_{m,p}$ such that

$$\mathbb{E} \left[\sup_{u, u' \in [s, t]} \|\zeta_{u \wedge T_m} - \zeta_{u' \wedge T_m}\|^p \right] \leq K_{m,p} |t - s|^{pH};$$

(ii) for $z \in \mathcal{K}_m$, $F(z, x) \equiv \mathbb{P}_\varepsilon(\mathcal{Y}(z, \varepsilon) \leq x)$ is continuous with density $f(z, x) \equiv \partial_x F(z, x)$ which is bounded away from zero in a neighborhood of $F^{-1}(z, \tau)$ and for x, x' in this neighborhood $|f(z, x) - f(z, x')| \leq K_m |x - x'|$. Moreover, $|f(z, x) - f(z', x)| \vee |F(z, x) - F(z', x)| \leq K_m \|z - z'\|$ for $z, z' \in \mathcal{K}_m$.

The state condition is the same as in [Assumption 2](#). Inverse CDF regularity implies that both $Q_t(\tau)$ and $f(\zeta_t, Q_t(\tau))$ the same smooth exponent H . The density conditions are local, so they do not exclude a squared-return density that is singular at zero but regular near a positive interior quantile. They do exclude atoms at the target quantile. [Bugni et al. \(2023\)](#) studies inference of distributional features with discrete observation in this model.

Assumption 5. *Let $\mathcal{H}_{n,i} \equiv \mathcal{F} \vee \sigma(\varepsilon_{n,k} : k \leq i)$ and $P_{n,i}(u, \tau) \equiv \mathbb{P}(\mathcal{Y}(\zeta_{i\Delta_n}, \varepsilon_{n,i}) - Q_{i\Delta_n}(\tau) \leq u \mid \mathcal{H}_{n,i-1})$. It holds for fixed $\tau \in (0, 1)$:*

⁶Sample quantiles have other applications, see, e.g., [Coeurjolly \(2008\)](#) for estimating the Hurst parameter of fractional Brownian motion using a convex combination of sample quantiles.

(i) almost surely, $P_{n,i}(0, \tau) = \tau$ and further there exist u_0 such that $\mathcal{H}_{n,i-1}$ -measurable $\pi_{n,i}(\cdot, \tau)$ such that $P_{n,i}(u, \tau) = \tau + \int_0^u \pi_{n,i}(v, \tau) dv$, $K^{-1} \leq \pi_{n,i}(u, \tau) \leq K$, and $|\pi_{n,i}(u, \tau) - \pi_{n,i}(0, \tau)| \leq K|u|$ for $|u| \leq u_0$;

(ii) let $f_t(x) \equiv f(\zeta_t, x)$, there exists a deterministic sequence d_n such that

$$\max_{1 \leq j \leq m_n} \left| \frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} \{ \pi_{n,i}(0, \tau) - f_{i\Delta_n}(Q_{i\Delta_n}(\tau)) \} \right| = O_p(d_n).$$

Condition (i) centers the quantile score at the specified threshold exactly, even after observing the disturbance history. It is not implied by a zero conditional mean of the raw noise. Because the score is a centered Bernoulli variable, its conditional variance is exactly $\tau(1 - \tau)$. In fact, its constant conditional Bernoulli law implies independence of the signs at this fixed quantile, given $\mathcal{F}$. The observations and neighboring-threshold indicators can still be dependent. Condition (ii) instead stabilizes the derivative of the predictable CDF: the block-average density at zero must approach the state-based density. It does not impose approximate quantile centering, which is already exact in (i). Under conditionally i.i.d. disturbances, density matching is observation-by-observation and $d_n = 0$, and disturbances with density f_ε such that $\sup_{x \in \mathcal{D}} (1 + \|x\|) f_\varepsilon(x) < \infty$ suffices in the location-scale model. Smooth heavy-tailed laws, including Cauchy disturbances, can therefore be admitted without observation moments.

Lemma 2 establishes uniform linearization over all blocks. Its proof adapts the classical localization-grid-inversion structure in Bahadur (1966) and Wu (2005), with Freedman inequalities (see, e.g., Freedman (1975)) to control short interval indicators. The union bound includes the growing block collection before Borel–Cantelli is applied. The almost-sure intrinsic remainder thus already contains the logarithmic cost of uniformity. The final statement below is in probability because density stabilization and the observation remainder are imposed in probability.

Lemma 2. Suppose Assumptions 1, 4, and 5 hold, $k_{n,j} \asymp \Delta_n^{-\varrho}$ uniformly for $1 \leq j \leq m_n$ such that $0 < \varrho < \frac{2H}{1+2H}$, $d_n = o(L_n^{-3/2})$, and $B_{n,R} = o(\Delta_n^{\varrho/2} L_n^{-1})$. Let $f_{n,j}(\tau) \equiv f_{\iota(1,j)\Delta_n}(Q_{\iota(1,j)\Delta_n}(\tau))$ and

$$U_{i,j}(\tau) \equiv \tau - \mathbb{1}\{\mathcal{Y}(\zeta_{\iota(i,j)\Delta_n}, \varepsilon_{n,\iota(i,j)}) \leq Q_{\iota(i,j)\Delta_n}(\tau)\}.$$

Then it holds for fixed $\tau \in (0, 1)$ that

$$\max_{1 \leq j \leq m_n} \sup_{t \in \mathcal{T}_{n,j}} \left| \sqrt{k_{n,j}} \{ \widehat{Q}_{n,t}(\tau) - Q_t(\tau) \} - \frac{1}{\sqrt{k_{n,j}}} \sum_{i=1}^{k_{n,j}} \frac{U_{i,j}(\tau)}{f_{n,j}(\tau)} \right| = o_p(L_n^{-1}).$$

The leading scores satisfy $|U_{i,j}(\tau)| \leq 1$, have conditional mean zero, and have exactly matched predictable variance $\tau(1 - \tau)$. Applying Lemma 1 chronologically therefore requires neither clipping nor a conditional skewness restriction. The resulting Gaussian target includes the derivative from the Bahadur representation.

Theorem 2. Under assumptions of Lemma 2, there exists a mixed Gaussian vector $Z_n = (Z_{n,j})_{1 \leq j \leq m_n} \in \mathbb{R}^{m_n}$ on an extension space with

$$Z_n \mid \mathcal{F} \sim \mathcal{N}\left(0, \text{diag}\left\{\frac{\tau(1-\tau)}{f_{n,j}(\tau)^2} : 1 \leq j \leq m_n\right\}\right),$$

such that

$$\max_{1 \leq j \leq m_n} \sup_{t \in \mathcal{T}_{n,j}} |\sqrt{k_{n,j}}\{\hat{Q}_{n,t}(\tau) - Q_t(\tau)\} - Z_{n,j}| = o_p(L_n^{-1}).$$

For feasible studentization, estimate the density of the state-based conditional distribution at its quantile, instead of the unconditional marginal density pooled over the observation horizon. The following rectangular-kernel estimator is local in time and centered at the estimated quantile.

Corollary 2. Under assumptions of Theorem 2, let $\text{cv}_{n,\alpha}$ be the $(1-\alpha)$ -quantile of the maximum of an $\mathbb{R}^{m_n}$ -dimensional folded standard Gaussian variable, for a deterministic bandwidth h_n with $h_n = o(L_n^{-1})$ and $h_n \Delta_n^{-\varrho/2} L_n^{-3/2} \rightarrow \infty$, define

$$\hat{f}_{n,j}(\tau) \equiv \Delta_n \vee \frac{1}{2h_n k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} \mathbb{1}\{|Y_{n,i} - \hat{Q}_{n,j}(\tau)| \leq h_n\}.$$

Then it holds for fixed $\tau \in (0, 1)$ that

$$\mathbb{P}\left(Q_t(\tau) \in \left[\hat{Q}_{n,t}(\tau) \pm \text{cv}_{n,\alpha} \frac{\sqrt{\tau(1-\tau)}}{\sqrt{k_{n,j}} \hat{f}_{n,j}(\tau)}\right] \text{ for all } t \in \mathcal{T}_{n,j}, 1 \leq j \leq m_n\right) = 1 - \alpha + o(1).$$

For polynomial bandwidths $h_n \asymp \Delta_n^\kappa$, the stated restrictions permit $0 < \kappa < \varrho/2$. The proof gives $\max_{1 \leq j \leq m_n} |\hat{f}_{n,j}/f_{n,j} - 1| = o_p(L_n^{-1})$, which is sufficiently accurate for studentized maximum inference. The admissible state and block-size range remains the same as for the mean. Density estimation can nevertheless matter in finite samples, motivating the inversion implementation below.

A separate coupling at each τ would not determine the cross-quantile dependence. To approximate the whole quantile curve, we impose a stronger and transparent independence condition. Exact score centering over an entire quantile interval would itself fix the conditional law of central rank processes, so it should not be presented as an innocuous extension of a single martingale score restriction.

Assumption 6. Conditional on $\mathcal{F}$, $(\varepsilon_{n,i})_{1 \leq i \leq n}$ are independent, hence $\mathcal{L}(\{\varepsilon_{n,i}\}_{1 \leq i \leq n} \mid \mathcal{F}) = \mathbb{P}_\varepsilon^{\otimes n}$.

Together with the marginal density regularity on the specified quantile interval, Assumption 6 implies Assumption 5 holds for all τ with $d_n = 0$. Fix $0 < \underline{\tau} < \bar{\tau} < 1$ and impose Assumption 4(ii) uniformly on this interval with a common neighborhood, conditional independence then makes the transformed variables i.i.d. uniforms. The uniform empirical-process coupling of Komlós et al. (1975) yields jointly independent Brownian bridges across blocks, with the correct within-block covariance across quantile levels. This is a different argument from repeating scalar

Strassen coupling. Related process-level approaches include Proposition 3.1 of [Cattaneo et al. \(2025\)](#) and Theorem 5 of [Belloni et al. \(2019\)](#). The target here is a sequence of conditional Gaussian processes indexed by (j, τ) .

Theorem 3. *Suppose Assumptions 1, 4, 6 hold, and $k_{n,j} \asymp \Delta_n^{-\varrho}$ uniformly for $1 \leq j \leq m_n$ such that $0 < \varrho < \frac{2H}{1+2H}$ and $B_{n,R} = o(\Delta_n^{\varrho/2} L_n^{-1})$. For $0 < \underline{\tau} < \bar{\tau} < 1$, there exist jointly conditionally Gaussian processes $\mathcal{Z}_{n,1}, \dots, \mathcal{Z}_{n,m_n}$ on an extension space such that*

$$\max_{1 \leq j \leq m_n} \sup_{\tau \in [\underline{\tau}, \bar{\tau}]} \sup_{t \in \mathcal{T}_{n,j}} |\sqrt{k_{n,j}} \{\widehat{Q}_{n,j}(\tau) - Q_t(\tau)\} - \mathcal{Z}_{n,j}(\tau)| = o_p(L_n^{-1}),$$

where $\mathcal{Z}_{n,j}(\tau) = B_{n,j}^0(\tau)/f_{n,j}(\tau)$ and conditionally on $\mathcal{F}$, $B_{n,j}^0$ are independent standard Brownian bridges.

The target is a growing collection of density-scaled bridges, not a single fixed process over calendar time. Its conditional covariance is

$$\text{Cov}(\mathcal{Z}_{n,j}(\tau), \mathcal{Z}_{n,l}(\tau') | \mathcal{F}) = \frac{\tau \wedge \tau' - \tau\tau'}{f_{n,j}(\tau)f_{n,j}(\tau')} \mathbb{1}\{j = l\}.$$

The compact interior interval prevents the density and quantile process normalization from degenerating at the endpoints. The theorem therefore provides joint control at nearby quantile levels. This gives a density-free band for a fixed target quantile by inverting those levels, avoiding the selection of bandwidth h_n in Corollary 2.

Corollary 3. *Under assumptions of Theorem 3, let $\text{cv}_{n,\alpha}$ be the $(1-\alpha)$ -quantile of the maximum of an $\mathbb{R}^{m_n}$ -dimensional folded standard Gaussian variable. Then it holds for fixed $\tau \in (\underline{\tau}, \bar{\tau})$ that*

$$\mathbb{P}\left(Q_t(\tau) \in \left[\widehat{Q}_{n,t}\left(\tau \pm \text{cv}_{n,\alpha} \sqrt{\frac{\tau(1-\tau)}{k_{n,j}}}\right)\right] \text{ for all } t \in \mathcal{T}_{n,j}, 1 \leq j \leq m_n\right) = 1 - \alpha + o(1).$$

The implementation follows the classical probability-scale inversion principle of [Woodruff \(1952\)](#). The process approximation in Theorem 3 provides the joint control required across blocks. The adjusted levels lie in the interior interval for all sufficiently small Δ_n . Their empirical endpoints use the same smallest minimizer convention as the estimator. This construction removes density estimation and permits asymmetric intervals on the observation scale.

3 Monte Carlo Simulation

We examine the finite-sample coverage of the proposed uniform confidence bands in six data-generating processes (DGPs) motivated by the three examples in Section 2.1: location-scale observations, Lévy-driven returns, and trading counts. The experiment covers conditional means and conditional medians, with the latter implemented by inversion of empirical quantile levels. Coverage is assessed jointly at all points of a fine numerical state grid, approximating simultaneous coverage over the observation period rather than average pointwise coverage. We report nominal coverage levels of 90% and 95%.

Table 2: State-process parameters

Setting	$\bar{\mu}_t$	ρ	η	α_t	κ	γ
A	1.2	8/252	1.25/252	1	5/252	$1/(4\sqrt{252})$
B	$1.2h(t)$	4/252	2.5/252	$h(t)$	4/252	$1/(2\sqrt{252})$

3.1 Simulation setting

Normalize the observation horizon to one trading day, $T = 1$. We use a dimensionless scale-squared process c , with the normalization absorbed into its dynamics and parameter values. The state variables are generated from

$$\begin{aligned} d\mu_t &= \rho(\bar{\mu}_t - \mu_t)dt + \eta dB_t, \\ dc_t &= \kappa(\alpha_t - c_t)dt + \gamma\sqrt{c_t}dB'_t, \end{aligned}$$

where B and B' are independent Brownian motions, independent of the observation innovations. Table 2 gives two parameter settings. Setting **A** has constant mean-reversion targets. Setting **B** doubles the diffusion coefficients, reduces the mean-reversion coefficients, and introduces a deterministic intraday pattern. Specifically, $h(t) = 1 + 0.1 \cos(2\pi t)$. This pattern is imposed on the targets, not directly on the realized paths. Initial values are $\mu_0 = \bar{\mu}_0$ and $c_0 = \alpha_0$. They are not drawn from stationary distributions. The settings provide controlled comparisons of state variation rather than an empirical calibration to a particular asset.

The first two designs correspond to the simple location-scale transformation

$$\begin{aligned} \text{DGP 1: } Y_{n,i} &= \mu_{i\Delta_n} + \varepsilon_{n,i}, & \varepsilon_{n,i} &\sim \mathcal{N}(0, 1), \\ \text{DGP 2: } Y_{n,i} &= \mu_{i\Delta_n} + \sqrt{c_{i\Delta_n}}\varepsilon_{n,i}, & \varepsilon_{n,i} &\sim \sqrt{14/16} \times t_{16}. \end{aligned}$$

The conditional mean is $G_t = \mu_t$ in both cases. The Student innovations have unit variance, making the observation noise scales comparable across the two designs, while introducing non-Gaussian tails and time-varying conditional variance in DGP 2. All observation innovations are independent of the state. The sixteen degrees of freedom permit moment orders strictly above twelve, sufficient for an appropriate finite-moment specialization of the theory.

For DGPs **3-5**, we generate zero-drift efficient log prices according to $P_t^{(\beta)} = \int_0^t \sqrt{c_s} dL_s^{(\beta)}$ and set $Y_{n,i} = \Delta_n^{-2/\beta} (P_{i\Delta_n}^{(\beta)} - P_{(i-1)\Delta_n}^{(\beta)})^2$. DGP **3** uses $\beta = 2$ and a standard Brownian motion $L^{(2)}$. DGPs **4** and **5** use symmetric stable processes with indices $\beta = 1.5$ and $\beta = 1$, respectively, normalized by $\mathbb{E}[\exp\{iu(L_{t+s}^{(\beta)} - L_t^{(\beta)})\}] = \exp\{-s|u|^\beta\}$. The target in DGP **3** is the normalized spot variance, $G_t = c_t$. In DGPs **4** and **5**, the target is instead the instantaneous conditional median $Q_t(1/2) = (F_\beta^{-1}(3/4))^2 c_t$, where F_β is the CDF of $L_1^{(\beta)}$ under the specified normalization. Numerically, $(F_{1.5}^{-1}(3/4))^2 \approx 0.9388$ and $(F_1^{-1}(3/4))^2 = 1$. Thus c_t is a scale-squared process, not a finite return variance, when $\beta < 2$. Dividing both endpoints of a median band by $(F_\beta^{-1}(3/4))^2$ gives a band for c_t , without changing its coverage.

DGP **6** is a high-intensity Cox specification. Conditional on the state, $N^{(n)}$ has intensity $\Delta_n^{-1}(\mu_t)_+$ in normalized time, where $x_+ = x \vee 0$, and

$$Y_{n,i} = N_{i\Delta_n}^{(n)} - N_{(i-1)\Delta_n}^{(n)}, \quad Y_{n,i} \mid \mathcal{F} \sim \text{Poisson}\left(\Delta_n^{-1} \int_{(i-1)\Delta_n}^{i\Delta_n} (\mu_s)_+ ds\right).$$

Counts are conditionally independent across disjoint intervals. The target is the normalized instantaneous intensity $G_t = (\mu_t)_+$, measured in trades per one-second interval at the sampling frequency used below. All generated μ paths are positive, so $G_t = \mu_t$ and the positive-part extension leaves every reported observation unchanged. We retain full counts rather than trade/no-trade indicators. This is a high-intensity experiment, not a fixed-intensity process observed on a finer grid.

We take $n = 23,400$ and $\Delta_n = 1/23,400$, corresponding to one-second observations during a trading day of 23,400 seconds. We consider equal block sizes $k_n \in \{300, 600, 1170\}$, giving $m_n \in \{78, 39, 20\}$ blocks. Their durations are 5, 10, and 19.5 minutes. Each block size divides n , so the partition uses the full observation period. Varying k_n changes both the precision of each local estimator and the amount of state variation that a blockwise constant band must accommodate.

Each DGP–parameter setting is evaluated using 5,000 independent Monte Carlo replications, regenerating both the state and the observation innovations. The same replication is used for all three block sizes. We use Euler discretization with four substeps per second for states. A replication is counted as covered only if the band contains the target at every numerical state-grid point in every block, including the horizon endpoints. The resulting criterion is a numerical approximation to continuum coverage. For mean targets we use the sample-variance studentization of Corollary **1**. For the median targets we use Corollary **3**, with endpoint levels $1/2 \pm \text{cv}_{n,\alpha}/(2\sqrt{k_n})$ hence the implementation requires no density bandwidth.

3.2 Results

Table **3** reports coverage percentages for all six DGPs and both parameter settings. Mean rows use Corollary **1**, whereas median rows use the inverted quantile levels in Corollary **3**. Thus the table evaluates the stated feasible implementation for each target, without using true conditional variances or densities.

The location-scale designs exhibit relatively small coverage distortions. Across both settings and all block sizes, coverage in DGPs **1** and **2** ranges from 87.54% to 89.90% at the 90% nominal level and from 93.24% to 95.14% at the 95% level. For example, with $k_n = 600$, setting **A** produces 88.90% and 89.88% coverage for the Gaussian and Student designs, respectively. The corresponding 95% bands cover 94.30% and 95.14% of replications. Modest undercoverage remains, but introducing time-varying conditional variance and Student innovations does not cause a marked deterioration.

The inverted median bands provide stable coverage under both heavy-tailed return designs. At $k_n = 600$, their 90% coverage rates range from 88.72% to 89.56%, and their 95% rates range

Table 3: Simultaneous coverage of conditional mean and median bands (%)

DGP	Setting	Nominal 90%			Nominal 95%		
		$k_n = 300$	$k_n = 600$	$k_n = 1170$	$k_n = 300$	$k_n = 600$	$k_n = 1170$
1	A	88.26	88.90	89.90	94.00	94.30	94.94
	B	87.54	88.26	88.44	93.24	93.96	94.18
2	A	88.76	89.88	89.40	94.18	95.14	94.76
	B	89.44	89.10	88.16	94.70	94.42	93.94
3	A	71.66	82.40	85.46	80.72	88.90	92.08
	B	70.78	80.94	82.90	80.52	88.30	89.84
4	A	89.30	89.56	89.12	95.32	95.18	94.26
	B	88.62	88.86	87.72	94.44	94.80	93.86
5	A	89.04	88.72	89.00	95.38	94.66	94.40
	B	88.36	88.74	87.52	94.90	95.08	94.08
6	A	87.08	87.94	88.62	92.76	93.92	94.08
	B	86.98	88.10	87.58	93.06	94.18	93.58

Note: Coverage requires success at every state grid point over the entire time span. 5,000 replications per DGP and setting. The respective numbers of blocks are 78, 39, and 20. DGPs **1** and **2** are Gaussian and Student location models. DGP **3** uses Brownian squared returns; DGPs **4** and **5** use stable-1.5 and Cauchy squared returns. DGP **6** uses high-intensity Cox counts. Targets are conditional means in DGPs **1-3** and **6** and conditional medians in DGPs **4** and **5**. Median endpoints use quantile-level inversion in Corollary 3, rather than the density-estimated bands of Corollary 2.

from 94.66% to 95.18%. Across all three block sizes, the corresponding ranges are 87.52-89.56% and 93.86-95.38%. The Cauchy design is not systematically less accurate than the stable-1.5 design, despite its heavier tails. These findings support conditional quantile inference when second moments of returns, and hence means of squared returns, do not exist. They do not represent an efficiency ranking of mean and median estimators, since their targets differ.

The Cox design also exhibits moderate undercoverage rather than severe failure. Its coverage ranges from 86.98% to 88.62% for 90% bands and from 92.76% to 94.18% for 95% bands. For setting **A**, increasing the block size improves coverage at both nominal levels. In setting **B**, the largest block does not perform best, consistent with the cost of approximating a more variable intensity path by a blockwise constant band.

The Brownian driven return design exhibits more pronounced finite-sample undercoverage than the other designs. At the 90% nominal level, coverage rises from 71.66% to 85.46% in setting **A** and from 70.78% to 82.90% in setting **B** as k_n increases from 300 to 1170. At the 95% level, the corresponding ranges are 80.72-92.08% and 80.52-89.84%. This is because Gaussian return innovations do not imply Gaussian squared-return observations: the latter are right-skewed, and a studentized simultaneous band depends on accurate approximation of

tail probabilities across many blocks. Coverage improves as the block size increases, although appreciable undercoverage remains at the largest block size considered. This motivates further finite-sample work. [Bollerslev et al. \(2021\)](#) also document similar results and propose to use fixed- k coupling to achieve better coverage.

It is also worth noting that larger blocks do not improve every design. In setting **B**, moving from $k_n = 600$ to 1170 lowers 90% median coverage from 88.86% to 87.72% in DGP **4** and from 88.74% to 87.52% in DGP **5**, consistent with the tradeoff between local precision and within-block state variation. Overall, the results show near-nominal coverage for location, counts, and quantile inversion band and modest mean band distortions for squared Lévy-driven returns.

4 Empirical illustration

We apply the proposed inference procedures to one-minute price and volume observations for the SPDR S&P 500 ETF Trust (SPY) and Bitcoin against the U.S. dollar (BTCUSD), obtained from [Barchart](#), on the eight scheduled FOMC announcement dates in 2025.⁷ All timestamps are aligned to New York local time, with the crypto source clock converted from Chicago time, and both assets are restricted to 09:30–16:00. Normalizing this interval to $T = 1$ gives $\Delta_n = 1/390$. We use thirteen nonoverlapping half-hour blocks with $k_{n,j} = 30$.⁸ SPY volatility is measured by the square root of the block mean of squared returns. For BTC, we use the square root of the normalized median of squared returns, a measure of typical return magnitude. Volume is the block mean of the vendor-reported one-minute field. We use the studentized mean bands for SPY squared returns and volume, and quantile-level inversion for BTC squared returns, applying a square root to the dispersion-band endpoints. Estimates are labeled by block start, and the lines connecting block values in the figures are visual guides rather than observations of intrablock pattern.

4.1 A second stage of equity-market adjustment

SPY exhibits a consistent second stage of adjustment during the press conference. To summarize its magnitude, [Table 4](#) compares the statement block, 14:00–14:30, and the first press-conference block, 14:30–15:00, with the immediately preceding block, 13:30–14:00. Each reported ratio is calculated separately for each announcement date and then summarized by its median across the eight dates. SPY volatility is 1.90 times its pre-announcement level in the statement block and 3.25 times that level in the conference block. The corresponding volume ratios are 2.32 and 4.04. Both quantities exceed their pre-announcement values during the conference block on all

⁷The dates are January 29, March 19, May 7, June 18, July 30, September 17, October 29, and December 10. The meeting calendar is available at <https://www.federalreserve.gov/monetarypolicy/fomccalendar.s.htm>.

⁸14:00–14:01 is excluded from both the price and volume calculations to reduce the influence of the initial announcement jump while preserving the clock-time partition.

Table 4: Changes around the FOMC statement and press conference

Quantity	Statement / pre 14:00 vs. 13:30	Conference / pre 14:30 vs. 13:30	Conference > pre Dates
SPY volatility	1.90	3.25	8/8
SPY reported volume	2.32	4.04	8/8
BTC price dispersion	1.53	1.37	7/8
BTC reported volume	1.01	1.03	4/8

Note: Entries in the two ratio columns are medians of date-specific ratios, not ratios of pooled estimates. Clock times denote the starts of half-hour blocks. The pre-announcement baseline is 13:30–14:00. The final column counts dates on which the conference-block point estimate strictly exceeds its pre-announcement value.

eight dates. Moreover, on every date, the conference-block estimates exceed their statement-block counterparts. The median of the daily conference-to-statement ratios is 1.35 for volatility and 1.45 for volume. These findings point to continued adjustment after the initial policy release, rather than an intraday response confined to one announcement minute.

The bands also distinguish several of the pre-announcement and conference-period features. For SPY volume, the lower endpoint of the conference-block band exceeds the upper endpoint of the pre-announcement band on all eight dates. The analogous separation occurs for volatility on five dates. September 17 is a particularly pronounced episode: conference-block volatility and volume are respectively 5.63 and 6.61 times their pre-announcement values. The point estimates and their reported simultaneous bands therefore give a common descriptive picture of the trading-activity increase.

This pattern complements the evidence of [Gómez-Cram and Grotteria \(2022\)](#), who link high-frequency price discovery on FOMC days to the content and timing of the subsequent press conference. Their analysis finds that discussion of changes in the newly released statement is associated with concentrated trading and price adjustment. [Bollerslev et al. \(2024a\)](#) also document related announcement and press-conference volatility patterns using range-based spot estimators and pointwise confidence intervals. The present analysis complements that evidence with simultaneous bands for both return dispersion and trading volume. The proposed uniform inference procedures describe the timing and magnitude of these changes in local distributional features.

4.2 Persistent trading activity and the market close

The volume profiles combine the announcement-related increases with a pronounced regular-session pattern. On all eight dates, SPY volume in the 13:30 block is below its opening-block level. Trading subsequently rises around the statement and press conference, but the largest volume estimate occurs in the final half hour, 15:30–16:00, on six dates. Across announcements,

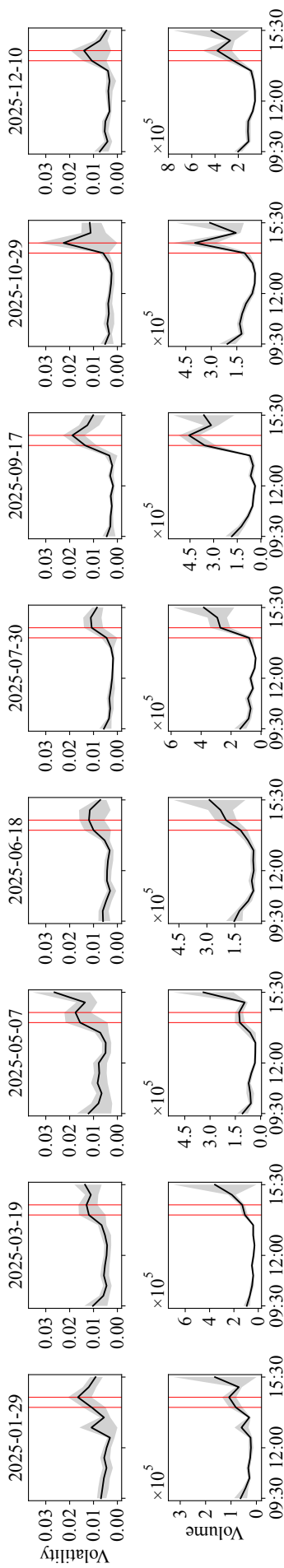


Figure 1: SPY volatility and trading volume on the 2025 FOMC days. Black lines display block estimates and gray shading displays the corresponding confidence bands. The first row reports volatility, and the second reports mean minute volume. Both use Corollary 1. Red lines mark the 14:00 statement and 14:30 press-conference start in New York time. Blocks are labeled by their start.

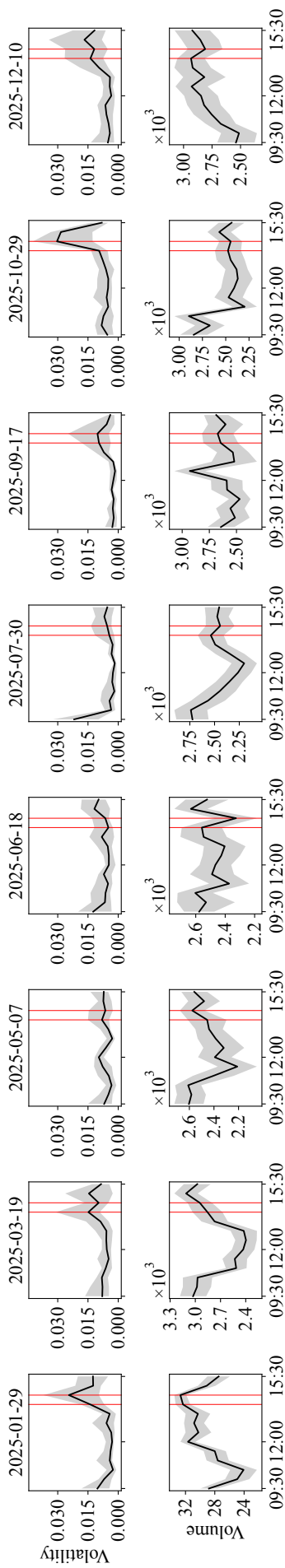


Figure 2: BTC return dispersion and reported trading volume on the 2025 FOMC days. Black lines display block estimates and gray shading displays the corresponding confidence bands. The first row reports median-based return dispersion and the second reports mean minute volume. Red lines mark 14:00 and 14:30 in New York time. Blocks are labeled by their start. Dispersion levels are not directly comparable with SPY's mean-based measure.

the median of the closing-to-pre volume ratios is 5.43, compared with ratios of 2.32 and 4.04 in the statement and conference blocks. The median of the daily closing-to-conference volume ratios is 1.43. Hence, the observed concentration of trading near the close exceeds that in the event windows on most sample days. This ordering describes the profiles; it does not separately identify the size of a diurnal effect. September 17 and October 29 are the exceptions: on those dates, the conference block itself has the largest volume estimate.

The timing of trading activity differs from that of volatility. [Bollerslev et al. \(2018\)](#) only study trading activity and volatility around public news. On six dates, SPY volatility in the final half hour (15:30–16:00) is lower than in the first conference block, even though trading remains elevated. The median closing-to-conference volatility ratio is 0.58. A natural interpretation is that trading continues as investors implement portfolio changes and complete orders, while the intensity of short-horizon price fluctuations has already subsided. This interpretation accords with [Fleming and Remolona \(1999\)](#), who document separate stages of price formation and trading after public-information releases in the Treasury market, with elevated volume persisting longer than elevated volatility.

The opening and closing concentration also connects to the established intraday volume pattern literature. [Jain and Joh \(1988\)](#) document systematic variation in equity trading volume across the trading day, while [Admati and Pfleiderer \(1988\)](#) show how the interaction of discretionary liquidity traders and informed traders can generate concentrated trading periods. In this light, the FOMC-related increase is superimposed on a strong trading-day rhythm rather than replacing it. The local estimates distinguish the renewed adjustment around 14:30 from the subsequent concentration of activity near the close. A single daily volume or volatility statistic would obscure this difference in timing.

4.3 Heterogeneous Bitcoin responses

BTC does not exhibit the same uniform response across announcement dates. Its median-based return dispersion rises in the conference block relative to the pre-announcement block on seven of the eight dates, but the median ratio is only 1.37 and masks considerable heterogeneity. The conference-to-pre ratio ranges from 0.78 on June 18 to 5.80 on January 29. March 19 shows only a modest increase, with a ratio of 1.10. By contrast, October 29 has a ratio of 3.94, and dispersion remains 3.69 times the pre-announcement level in the following 15:00–15:30 block. For January 29 and October 29, the conference-block lower band exceeds the pre-announcement upper band. On October 29 the separation also persists into the 15:00 block. Because the BTC statistic is based on a median, these episodes concern an increase in typical absolute one-minute returns, not merely the influence of a single large squared-return observation.

This event heterogeneity is relevant to the differing assessments of Bitcoin’s sensitivity to monetary news. [Karau \(2023\)](#) documents a relationship between monetary policy and Bitcoin, including elevated volatility around monetary-policy announcements, whereas [Benigno and Rosa \(2023\)](#) find little systematic response of Bitcoin prices to monetary and macroeco-

nomic news in their event-study sample. Our results focus on local return magnitude rather than the direction of the price response. Small or offsetting signed returns can coexist with substantial changes in dispersion. Examining the intraday conditional distribution therefore adds information beyond a pooled signed-return response: several announcements coincide with only modest changes, while selected dates display a much more pronounced and persistent adjustment.

The vendor-reported BTC volume series has a different within-day pattern from SPY’s; we compare date-specific ratios rather than pool their levels across months. Its median statement-to-pre and conference-to-pre ratios are 1.01 and 1.03, respectively, and conference-block volume exceeds the pre-announcement value on only four dates. On October 29, for example, the pre-announcement and conference blocks have the same reported mean volume even though the median-based dispersion measure rises almost fourfold. There is also no closing-volume peak in the selected 15:30–16:00 window on any of the eight BTC dates. Thus the supplied feed records a much smaller accompanying volume increase than in the equity example. This is a statement about the recorded field, not verified market-wide Bitcoin turnover.

The market’s trading schedule provides a useful perspective on this contrast. Bitcoin trades 24 hours a day. The 09:30–16:00 restriction aligns the comparison with U.S. market hours; it does not define BTC’s trading day. Continuous trading does not, however, imply constant activity: [Hansen et al. \(2024\)](#) document systematic hourly and weekly patterns in cryptocurrency volatility and volume, as well as differences in price discovery across trading venues. The absence of an equity-style closing peak in our selected window is consistent with a different intraday organization of liquidity. News-related repricing and changes in available liquidity are possible explanations for large movements in typical returns without a commensurate rise in reported trading volume.

Taken together, the illustration shows why local inference for several distributional features is useful: SPY has a consistent conference stage increase followed by persistent trading near the close, whereas BTC combines modest responses on some dates with large changes in typical return magnitude on others. These comparisons describe eight announcement days and do not separately identify statement, conference, or diurnal effects. A non-announcement benchmark, as in [Bollerslev et al. \(2018\)](#), would be needed for that decomposition.

5 Concluding Remarks

We develop uniform inference for conditional mean and quantile processes of high-frequency observations. The main approximation argument exploits chronologically ordered scalar blocks to avoid the polynomial dimension cost of a direct general vector coupling. This permits local sample sizes that are compatible with rough states. A block-uniform Bahadur representation extends the method to fixed conditional quantiles, while conditional independence permits a joint Brownian-bridge approximation over an interior quantile interval and a density-free

inversion procedure.

The simulation results show good finite-sample performance. The empirical illustration shows the value of reporting the whole pattern of intraday estimates: equity-market adjustment continues during the FOMC press conference, trading remains concentrated near the close, and Bitcoin’s typical return magnitude responds heterogeneously across announcements. These are different features of market adjustment that daily aggregates and isolated pointwise estimates do not reveal.

Several extensions merit further work. The present results provide a common inferential structure for these questions, while making explicit the additional observation and dependence conditions that each requires. Data-driven block selection and finite-sample refinements for skewed moment estimators are natural extensions. The fixed- k approaches of [Bollerslev et al. \(2021\)](#) and [Bollerslev et al. \(2024b\)](#) offer a complementary starting point for the latter.

Appendix: Proofs

Throughout the proof, we use K and K' to denote some positive constants that may change from line to line, and write K_p to emphasize its dependence on some parameter p . For vector x , we write $[x]_a$ as its a th coordinate.

Proof of Theorem 1. By the standard localization argument in Section 4.4.1 of [Jacod and Protter \(2012\)](#), it suffices to prove the theorem under the strengthened conditions that Assumption 2 holds with $T_1 = \infty$, the state takes values in compact set $\mathcal{K}$, and the assumed bounds hold with deterministic constants K and K_p .

We start with the following decomposition

$$\begin{aligned} \sqrt{k_{n,j}}(\widehat{G}_{n,j} - G_t) - Z_{n,j} &= \widetilde{Z}_{n,j} - \bar{\sigma}_{n,j}\xi_{n,j} + (\bar{\sigma}_{n,j} - \sigma_{\iota(1,j)\Delta_n})\xi_{n,j} \\ &\quad + \sqrt{k_{n,j}}(\bar{G}_{n,j} - G_t) + \frac{1}{\sqrt{k_{n,j}}} \sum_{i \in \mathcal{I}_{n,j}} R_{n,i} \\ &\equiv \widetilde{Z}_{n,j} - \bar{\sigma}_{n,j}\xi_{n,j} + (I) + (II) + (III), \end{aligned} \tag{A.1}$$

where $\xi_{n,j}$ is constructed below using Lemma 1 with the conditional Strassen theorem in [Cattaneo et al. \(2025\)](#), and

$$\widetilde{Z}_{n,j} \equiv \frac{1}{\sqrt{k_{n,j}}} \sum_{i \in \mathcal{I}_{n,j}} U_{n,i}, \quad \bar{G}_{n,j} \equiv \frac{1}{k_{n,j}} \sum_{i=1}^{k_{n,j}} G_{\iota(i,j)\Delta_n}, \quad \bar{\sigma}_{n,j}^2 \equiv \frac{1}{k_{n,j}} \sum_{i=1}^{k_{n,j}} \sigma_{\iota(i,j)\Delta_n}^2.$$

We first construct a truncated version of $U_{n,i}$ by stopping at the first exit time with an auxiliary variance repair to apply Lemma 1. Let $b \geq 1$ be deterministic, define the first exit time $\tau(b) \equiv \inf\{1 \leq i \leq n : |U_{n,i}| > b\}$ and $J_i(b) \equiv \mathbb{1}\{\tau(b) \geq i\}$ where $\inf(\emptyset) = n+1$. Further denote

$$\begin{aligned} v_i &\equiv \mathbb{E}[U_{n,i}^2 \mid \mathcal{H}_{n,i-1}], \quad a_i \equiv \mathbb{E}[U_{n,i} \mathbb{1}\{|U_{n,i}| > b\} \mid \mathcal{H}_{n,i-1}], \\ w_i &\equiv \mathbb{E}[U_{n,i}^2 \mathbb{1}\{|U_{n,i}| > b\} \mid \mathcal{H}_{n,i-1}] + a_i^2. \end{aligned}$$

By definition and Assumption 3, $\mathbb{E}[U_{n,i} \mathbb{1}\{|U_{n,i}| \leq b\} \mid \mathcal{H}_{n,i-1}] = -a_i$ and $\text{Var}(U_{n,i} \mathbb{1}\{|U_{n,i}| \leq b\} \mid \mathcal{H}_{n,i-1}) = v_i - w_i$. On an extension equipped with independent Rademacher variables $(\psi_i)_{1 \leq i \leq n}$ which are independent of all original variables, let

$$U_{n,i}^b \equiv J_i(b)(U_{n,i} \mathbb{1}\{|U_{n,i}| \leq b\} + a_i + \sqrt{w_i}\psi_i) + (1 - J_i(b))\sqrt{v_i}\psi_i. \tag{A.2}$$

Let $\mathcal{H}'_{n,i} \equiv \mathcal{H}_{n,i} \vee \sigma(\psi_1, \dots, \psi_i)$, we have

$$\mathbb{E}[U_{n,i}^b \mid \mathcal{H}'_{n,i-1}] = 0, \quad \mathbb{E}[(U_{n,i}^b)^2 \mid \mathcal{H}'_{n,i-1}] = v_i, \quad |U_{n,i}^b| \leq 2b + \sqrt{K} \leq K'b.$$

Moreover, it follows Assumption 3 that

$$\left| \frac{1}{k_{n,j}\bar{\sigma}_{n,j}^2} \sum_{i \in \mathcal{I}_{n,j}} v_i - 1 \right| \leq Kd_n.$$

Thus the standardized auxiliary increments, $U_{n,i}^b/(\bar{\sigma}_{n,j}\sqrt{k_{n,j}})$, satisfies all requirements in Lemma 1, with every original predictable variance preserved.

For a uniform bound on $|U_{n,i}^b - U_{n,i}|$, let $M_n \equiv \max_{1 \leq i \leq n} |U_{n,i}|$, we have for $q = 1, 2$ that

$$\begin{aligned} \mathbb{E} \left[\sum_{i=1}^n J_i(b) \mathbb{E}[|U_{n,i}|^q \mathbb{1}\{|U_{n,i}| > b\} \mid \mathcal{H}_{n,i-1}] \right] &= \mathbb{E} \left[\sum_{i=1}^n J_i(b) |U_{n,i}|^q \mathbb{1}\{|U_{n,i}| > b\} \right] \\ &= \mathbb{E}[|U_{n,\tau(b)}|^q \mathbb{1}\{\tau(b) \leq n\}] \leq \mathbb{E}[M_n^q \mathbb{1}\{M_n > b\}]. \end{aligned} \quad (\text{A.3})$$

It follows Markov's inequality that $\mathbb{E}[M_n^q \mathbb{1}\{M_n > b\}] \leq K(B_{n,1}b^{q-3} \wedge B_{n,2}b^{q-4})$. Therefore,

$$\mathbb{P}(\tau(b) \leq n) \leq K(B_{n,1}b^{-3} \wedge B_{n,2}b^{-4}). \quad (\text{A.4})$$

On event $\{\tau(b) > n\}$, we have $U_{n,i}^b - U_{n,i} = a_i + \sqrt{w_i}\psi_i$. For the first term, it follows (A.3) that $\mathbb{E}[\sum_{i=1}^n J_i(b)a_i^2] \leq K(B_{n,1}b^{-1} \wedge B_{n,2}b^{-2})$ and $\mathbb{E}[\sum_{i=1}^n J_i(b)|a_i|] \leq K(B_{n,1}b^{-2} \wedge B_{n,2}b^{-3})$. For the second term, independence of ψ_i gives

$$\mathbb{E} \left[\sum_{j=1}^{m_n} \left| \frac{1}{\sqrt{k_{n,j}}} \sum_{i \in \mathcal{I}_{n,j}} J_i(b) \sqrt{w_i} \psi_i \right|^2 \right] \leq K \Delta_n^e \mathbb{E} \left[\sum_{i=1}^n J_i(b) w_i \right] \leq K \Delta_n^e (B_{n,1}b^{-1} \wedge B_{n,2}b^{-2}).$$

Combining with (A.4), if $(B_{n,1}b^{-3} \wedge B_{n,2}b^{-4}) = o(1)$, we can conclude

$$\begin{aligned} \max_{1 \leq j \leq m_n} \left| \frac{1}{\sqrt{k_{n,j}}} \sum_{i \in \mathcal{I}_{n,j}} (U_{n,i}^b - U_{n,i}) \right| &= O_p(b \Delta_n^{e/2}), \\ \max_{1 \leq j \leq m_n} \frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} (U_{n,i}^b - U_{n,i})^2 &= O_p(b^2 \Delta_n^e). \end{aligned} \quad (\text{A.5})$$

For the required clipping rate, we take $b = b_n = \Delta_n^{-e/12} L_n^{-5/4} (B_{n,1}^{1/6} \wedge B_{n,2}^{1/8})$ in (A.2), which satisfies $(B_{n,1}b_n^{-3} \wedge B_{n,2}b_n^{-4}) = o(1)$. Moreover, the rate condition in Theorem 1 implies the first bound in (A.5) is $o_p(L_n^{-1})$ and the second is $o_p(L_n^{-2})$. Denote

$$W_{n,j} = \frac{\tilde{Z}_{n,j}^b}{\bar{\sigma}_{n,j}} \equiv \frac{1}{\sqrt{k_{n,j}}} \sum_{i \in \mathcal{I}_{n,j}} \frac{U_{n,i}^b}{\bar{\sigma}_{n,j}}, \quad k_n \equiv \min_{1 \leq j \leq m_n} k_{n,j}, \quad k_n^* \equiv \max_{1 \leq j \leq m_n} k_{n,j}.$$

We construct the coupling variables starting from the first block. In particular, at block j , apply Lemma 1 to the sequence $\{U_{n,i}^b/(\bar{\sigma}_{n,j}\sqrt{k_{n,j}})\}_{i \in \mathcal{I}_{n,j}}$ with filtration $\mathcal{A}_0 = \mathcal{H}'_{n,\ell(1,j)-1}$, $\mathcal{A}_i = \mathcal{H}'_{n,\ell(i,j)}$ for $1 \leq i \leq k_{n,j}$, set $\ell = L_n$, and

$$\eta = K_C \left\{ \left(\frac{(b_n^3 + L_n^{3/2}) L_n^{7/2}}{\sqrt{k_n}} \right)^{1/4} + (d_n L_n^2)^{1/3} \right\},$$

where by the rate condition we have

$$\frac{(b_n^3 + L_n^{3/2}) L_n^{7/2}}{\sqrt{k_n} L_n^{-4}} \leq \left(\frac{B_{n,1}^{2/3} \wedge B_{n,2}^{1/2}}{k_n^{1/3} L_n^{-5}} \right)^{3/4} + \frac{L_n^9}{\sqrt{k_n}} = o(1), \quad \frac{d_n L_n^2}{L_n^{-3}} = d_n L_n^5 = o(1).$$

Therefore, $\eta = o(L_n^{-1})$. At each block, enlarge the filtration by all previously used coupling uniforms when applying Lemma 1. These uniforms are independent of the original variables

and Rademachers, so the martingale and variance conditions are unchanged. For conditional Strassen, consider independent uniform variables $V_{n,j}$ independent of all original variables and ψ_i 's, denote $\mathcal{H}_{n,j}'' \equiv \mathcal{H}_{n,\iota(1,j)-1}' \vee \sigma(V_{n,1}, \dots, V_{n,j-1})$. It follows Lemma SA.1 in Cattaneo et al. (2025) that there exists $\xi_{n,j}$ on $\mathcal{H}_{n,j}'' \vee \sigma(W_{n,j}, V_{n,j})$ such that

$$\xi_{n,j} \mid \mathcal{H}_{n,j}'' \sim \mathcal{N}(0, 1), \quad \mathbb{P}(|W_{n,j} - \xi_{n,j}| > \eta) \leq 2e^{-L_n} = 2\Delta_n, \quad (\text{A.6})$$

and every earlier $\xi_{n,l}$ for $l < j$ is measurable with respect to $\mathcal{H}_{n,j}''$. Iterated conditioning now gives, for arbitrary real $x_1, \dots, x_{m_n}$,

$$\mathbb{E} \left[\prod_{j=1}^{m_n} \mathbb{1}\{\xi_{n,j} \leq x_j\} \mid \mathcal{F} \right] = \Phi(x_{m_n}) \mathbb{E} \left[\prod_{j=1}^{m_n-1} \mathbb{1}\{\xi_{n,j} \leq x_j\} \mid \mathcal{F} \right] = \prod_{j=1}^{m_n} \Phi(x_j),$$

thus $(\xi_{n,j})_{1 \leq j \leq m_n} \mid \mathcal{F} \sim \mathcal{N}(0, I_{m_n})$, setting $Z_{n,j} = \sigma_{\iota(1,j)\Delta_n} \xi_{n,j}$ gives the required coupling variable Z_n . Note that the error bound in (A.6) is unconditional, hence

$$\mathbb{P} \left(\max_{1 \leq j \leq m_n} |W_{n,j} - \xi_{n,j}| > \eta \right) \leq 2m_n \Delta_n = o(1).$$

Since $\bar{\sigma}_{n,j} \leq K$ and $\eta = o(L_n^{-1})$, this implies $\max_{1 \leq j \leq m_n} |\tilde{Z}_{n,j}^b - \bar{\sigma}_{n,j} \xi_{n,j}| = o_p(L_n^{-1})$. Combining with (A.5) yields

$$\max_{1 \leq j \leq m_n} |\tilde{Z}_{n,j} - \bar{\sigma}_{n,j} \xi_{n,j}| = o_p(L_n^{-1}). \quad (\text{A.7})$$

We next turn to part (II). It follows from Assumption 2 that both $g(\cdot)$ and $\sigma(\cdot)$ are Lipschitz. Therefore, by the maximal inequality, we have

$$\begin{aligned} & \mathbb{E} \left[\left\{ \max_{1 \leq j \leq m_n} \sqrt{k_{n,j}} \sup_{t \in \mathcal{T}_{n,j}} |\bar{G}_{n,j} - G_t| \right\}^p \right] \\ & \leq K \sum_{j=1}^{m_n} k_{n,j}^{p/2} \mathbb{E} \left[\sup_{u, u' \in \mathcal{T}_{n,j}} \|\zeta_u - \zeta_{u'}\|^p \right] \\ & \leq K m_n (k_n^*)^{p/2} (k_n^* \Delta_n)^{pH} \leq K' \Delta_n^{(pH-1)(1-\varrho)-p\varrho/2}. \end{aligned} \quad (\text{A.8})$$

Taking $p > \frac{2(1-\varrho)}{2H(1-\varrho)-\varrho}$ makes the right-hand side $o(L_n^{-p})$.

For part (I), note that $(\bar{\sigma}_{n,j} - \sigma_{\iota(1,j)\Delta_n})$ is $\mathcal{F}$ -measurable and $\mathbb{E}[|\xi_{n,j}|^p \mid \mathcal{F}] = \mathbb{E}[|\mathcal{N}(0, 1)|^p] \leq K$, therefore by iterated expectation

$$\begin{aligned} & \mathbb{E} \left[\max_{1 \leq j \leq m_n} |(\bar{\sigma}_{n,j} - \sigma_{\iota(1,j)\Delta_n}) \xi_{n,j}|^p \right] \\ & \leq K \sum_{j=1}^{m_n} \mathbb{E} \left[\sup_{u, u' \in \mathcal{T}_{n,j}} \|\zeta_u - \zeta_{u'}\|^p \right] \\ & \leq K m_n (k_n^* \Delta_n)^{pH} \leq K' \Delta_n^{(pH-1)(1-\varrho)}. \end{aligned} \quad (\text{A.9})$$

Taking $p > 1/H$ makes the right-hand side $o(L_n^{-p})$.

For part (III), by $B_{n,R} = o(\Delta_n^{\varrho/2} L_n^{-1})$ we have

$$\max_{1 \leq j \leq m_n} \left| \frac{1}{\sqrt{k_{n,j}}} \sum_{i \in \mathcal{I}_{n,j}} R_{n,i} \right| \leq \sqrt{k_n^*} \max_{1 \leq i \leq n} |R_{n,i}| = O_p(\Delta_n^{-\varrho/2} B_{n,R}) = o_p(L_n^{-1}). \quad (\text{A.10})$$

The proof is completed by (A.1), (A.7)–(A.10). Q.E.D.

Proof of Lemma 1. Throughout this proof, we work under a regular conditional law given $\mathcal{A}_0$ and suppress this conditioning in expectations.

For integer $r \geq 2$, let $\ell_r \equiv \log(2r)$. Make r independent copies of entire filtered martingale block $(U_i)_{1 \leq i \leq k}$ under this conditional law and write $X_i \equiv (U_i^{(1)}, \dots, U_i^{(r)})$. Under a standard product filtration, X_i is a vector martingale difference with diagonal predictable covariance Q_i . Note that $|\sum_{i=1}^k \mathbb{E}[U_i^2 | \mathcal{A}_{i-1}] - 1| \leq Cd$ implies $(1 - Cd)I_r \leq \sum_{i=1}^k Q_i \leq (1 + Cd)I_r$ a.s. Adapting Parts 1 and 3 of the proof of Lemma SA.8 in [Cattaneo et al. \(2025\)](#) that for independent standard Gaussian vectors N , N_i , and bounded smooth $f(\cdot)$ that

$$\left| \mathbb{E} \left[f \left(\sum_{i=1}^k X_i \right) \right] - \mathbb{E}[f(N)] \right| \leq \frac{D_3(f)}{6} \sum_{i=1}^k \mathbb{E}[\|X_i\|_\infty^3 + \|Q_i^{1/2} N_i\|_\infty^3] + \frac{3}{2} Cd D_2(f),$$

where $D_q(f) \equiv \sup_{x \in \mathbb{R}^r} \sum_{a_1, \dots, a_q=1}^r |\partial_{a_1 \dots a_q} f(x)|$. We adopt the smooth maxima used in the proof of Theorem 4.1 in [Chernozhukov et al. \(2014\)](#), namely, take

$$f_{x,h}(z) \equiv g\left(\frac{s_\beta(z) - x}{h}\right), \quad s_\beta(z) \equiv \frac{1}{\beta} \log\left(\sum_{a=1}^r e^{\beta[z]_a}\right),$$

where $\beta = \ell_r/h$ and $g(\cdot)$ is a fixed nonincreasing C^3 cutoff, equal to one on $(-\infty, 0]$ and zero on $[1, \infty)$. Lemma 4.3 of [Chernozhukov et al. \(2014\)](#) gives $D_2(f_{x,h}) \leq Kh^{-2}\ell_r$ and $D_3(f_{x,h}) \leq Kh^{-3}\ell_r^2$, hence

$$\left| \mathbb{E} \left[f_{x,h} \left(\sum_{i=1}^k X_i \right) \right] - \mathbb{E}[f_{x,h}(N)] \right| \leq K_C \left\{ \frac{(b^3 + \ell_r^{3/2})\ell_r^2}{\sqrt{k}h^3} + \frac{d\ell_r}{h^2} \right\}, \quad (\text{A.11})$$

where the first term uses the Gaussian tail bound $\sum_{i=1}^k \mathbb{E}[\|Q_i^{1/2} N_i\|_\infty^3] \leq K\ell_r^{3/2}/\sqrt{k}$. By construction of $f_{x,h}(\cdot)$, we have for any $w \in \mathbb{R}^r$ that

$$\mathbb{1} \left\{ \max_{1 \leq a \leq r} [w]_a \leq x \right\} \leq f_{x+h,h}(w) \leq \mathbb{1} \left\{ \max_{1 \leq a \leq r} [w]_a \leq x + 2h \right\}.$$

Denote $F(\cdot)$ the $\mathcal{A}_0$ -conditional distribution of W and $\Phi(\cdot)$ the distribution of standard Gaussian variable, then $\mathbb{P}(\max_{1 \leq a \leq r} \sum_{i=1}^k U_i^{(a)} \leq x | \mathcal{A}_0) = F(x)^r$ and $\mathbb{P}(\max_{1 \leq a \leq r} [N]_a \leq x) = \Phi(x)^r$. Combining with (A.11) gives

$$F(x)^r \leq \mathbb{E} \left[f_{x+h,h} \left(\sum_{i=1}^k X_i \right) \right] \leq \Phi(x + 2h)^r + K_C \left\{ \frac{(b^3 + \ell_r^{3/2})\ell_r^2}{\sqrt{k}h^3} + \frac{d\ell_r}{h^2} \right\}. \quad (\text{A.12})$$

Similarly, we can show

$$F(x)^r \geq \mathbb{E} \left[f_{x-h,h} \left(\sum_{i=1}^k X_i \right) \right] \geq \Phi(x - 2h)^r - K_C \left\{ \frac{(b^3 + \ell_r^{3/2})\ell_r^2}{\sqrt{k}h^3} + \frac{d\ell_r}{h^2} \right\}. \quad (\text{A.13})$$

Moreover, by the Gaussian anti-concentration inequality (see, e.g., Lemma A.1 in [Chernozhukov et al. \(2017\)](#)), we have

$$\sup_{x \in \mathbb{R}} \{ \Phi(x + 2h)^r - \Phi(x - 2h)^r \} \leq K_C h \sqrt{\ell_r}. \quad (\text{A.14})$$

Combining (A.12)–(A.14) then gives

$$\sup_{x \in \mathbb{R}} |F(x)^r - \Phi(x)^r| \leq K_C \left\{ h\sqrt{\ell_r} + \frac{(b^3 + \ell_r^{3/2})\ell_r^2}{\sqrt{k}h^3} + \frac{d\ell_r}{h^2} \right\}.$$

To optimize, take $h = (b^3 + \ell_r^{3/2})^{1/4} \ell_r^{3/8} k^{-1/8} + d^{1/3} \ell_r^{1/6}$. Thus

$$\sup_{x \in \mathbb{R}} |F(x)^r - \Phi(x)^r| \leq K_C \left\{ \left(\frac{(b^3 + \ell_r^{3/2})\ell_r^{7/2}}{\sqrt{k}} \right)^{1/4} + (d\ell_r^2)^{1/3} \right\}. \quad (\text{A.15})$$

Now, fix the level ℓ in the desired statement, and for simplicity denote $\epsilon \equiv \eta/K_C$. For $x \geq 0$, set $r_x = \lceil 1/(1 - \Phi(x)) \rceil$, then we can verify $1 \leq r_x(1 - \Phi(x)) \leq 3/2$, $e^{-3} \leq \Phi(x)^{r_x} \leq e^{-1}$, and $\log(2r_x) \leq K(1 + x^2)$. For $x \in [0, 1 + \sqrt{2\ell}]$, $\log(2r_x) \leq K\ell$, so right-hand side of (A.15) is at most $K'\epsilon$. Under the restriction $\eta \leq 1$, choose K_C sufficiently large that that $K'\epsilon \leq e^{-3}/2$. The triangle inequality then gives $F(x)^{r_x} \geq \Phi(x)^{r_x} - K'\epsilon \geq e^{-3}/2$. By the mean value theorem, we have

$$|F(x)^{r_x} - \Phi(x)^{r_x}| \geq r_x[F(x) \wedge \Phi(x)]^{r_x-1} |F(x) - \Phi(x)| \geq \frac{e^{-3}}{2} r_x |F(x) - \Phi(x)|.$$

Since $r_x^{-1} \leq 1 - \Phi(x)$, this gives $|F(x) - \Phi(x)| \leq K\epsilon(1 - \Phi(x))$. Reflection on $x \in [-1 - \sqrt{2\ell}, 0]$ yields

$$|F(x) - \Phi(x)| \leq K\epsilon[\Phi(x) \wedge (1 - \Phi(x))], \quad \text{for } |x| \leq 1 + \sqrt{2\ell}. \quad (\text{A.16})$$

The same bound holds with $F(x-)$ in place of $F(x)$. To convert relative errors into bounds on η -shifts, set an auxiliary function $J(t) \equiv \log(t/(1-t))$. Note that if $|a-t| \leq v(t \wedge (1-t))$ with $v \leq 1/2$, both a/t and $(1-a)/(1-t)$ lie between $1-v$ and $1+v$, hence $|J(a) - J(t)| \leq 4v$. Moreover, since $1 - \Phi(x) \leq \phi(x)/x$ for $x \geq 1$, then

$$\frac{d}{dx} J(\Phi(x)) = \frac{\phi(x)}{\Phi(x)(1 - \Phi(x))} \geq K > 0.$$

Recall $\eta \leq 1$, integrating above derivative over $(x - \eta, x)$ and $(x, x + \eta)$, combining with (A.16) yields

$$\Phi(x - \eta) \leq F(x-) \leq F(x) \leq \Phi(x + \eta) \quad \text{for } |x| \leq \sqrt{2\ell} + \eta. \quad (\text{A.17})$$

Let $I \equiv [-\sqrt{2\ell} - \eta, \sqrt{2\ell} + \eta]$. For any compact set $D \subset I$, there exist finitely many intervals $[a_n - \eta, b_n + \eta]$ that are disjoint except for boundaries such that $D \subset \bigcup_n [a_n, b_n]$ and $D^\eta = \bigcup_n [a_n - \eta, b_n + \eta]$, hence (A.17) yields

$$\mathbb{P}(W \in D \mid \mathcal{A}_0) \leq \sum_n [F(b_n) - F(a_n-)] \leq \sum_n [\Phi(b_n + \eta) - \Phi(a_n - \eta)] = \nu(D^\eta \mid \mathcal{A}_0).$$

Therefore, for any $B \in \mathcal{B}(\mathbb{R})$, by inner regularity we have

$$\mathbb{P}(W \in B \cap I \mid \mathcal{A}_0) = \sup_D \mathbb{P}(W \in D \mid \mathcal{A}_0) \leq \nu(B^\eta \mid \mathcal{A}_0), \quad (\text{A.18})$$

where the supremum is taken over all compact $D \subset B \cap I$ and the last step is by $D^\eta \subset B^\eta$. Moreover, from (A.17) we have

$$\mathbb{P}(W \notin I \mid \mathcal{A}_0) = F((-\sqrt{2\ell} - \eta)-) + 1 - F(\sqrt{2\ell} + \eta) \leq 2(1 - \Phi(\sqrt{2\ell})) \leq 2e^{-\ell}.$$

Combining with (A.18) completes the proof. Q.E.D.

Proof of Corollary 1. We adopt the notation used in the proof of Theorem 1. Recall that $|U_{n,i}^b| \leq Kb$ and $\mathbb{E}[(U_{n,i}^b)^2 | \mathcal{H}'_{n,i-1}] = v_i \leq K$. By Freedman's inequality (see Theorem 1.6 in Freedman (1975)), taking $x = 2L_n$ yields

$$\mathbb{P}\left(\left|\frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} U_{n,i}^b\right| > \sqrt{\frac{4KL_n}{k_n}} + \frac{4KbL_n}{k_n}\right) \leq K'\Delta_n^2.$$

Since $m_n\Delta_n^2 = o(1)$, a union bound over $1 \leq j \leq m_n$ then gives

$$\max_{1 \leq j \leq m_n} \left|\frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} U_{n,i}^b\right| = O_p\left(\sqrt{\frac{L_n}{k_n}} + b\frac{L_n}{k_n}\right). \quad (\text{A.19})$$

Note that $|(U_{n,i}^b)^2 - v_i| \leq Kb^2$ and $\mathbb{E}[(U_{n,i}^b)^2 - v_i]^2 | \mathcal{H}'_{n,i-1}] \leq Kb^2\mathbb{E}[(U_{n,i}^b)^2 | \mathcal{H}'_{n,i-1}] \leq K'b^2$, with a similar argument as deriving (A.19) we can show

$$\max_{1 \leq j \leq m_n} \left|\frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} \{(U_{n,i}^b)^2 - v_i\}\right| = O_p\left(b\sqrt{\frac{L_n}{k_n}} + b^2\frac{L_n}{k_n}\right). \quad (\text{A.20})$$

For variable V , denote the blockwise variance operator as $s_j^2(V) \equiv k_{n,j}^{-1} \sum_{i \in \mathcal{I}_{n,j}} (V_i - \bar{V}_j)^2$. It follows (A.5) that

$$\max_{1 \leq j \leq m_n} |s_j(U) - s_j(U^b)| \leq \max_{1 \leq j \leq m_n} \left\{ \frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} (U_{n,i} - U_{n,i}^b)^2 \right\}^{1/2} = O_p(b\Delta_n^{q/2}). \quad (\text{A.21})$$

Assumption 2, 3, and the maximal inequality yield

$$\max_{1 \leq j \leq m_n} \left|\frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} v_i - \sigma_{\iota(1,j)\Delta_n}^2\right| = O_p(d_n + m_n^{1/p}(k_n^*\Delta_n)^H). \quad (\text{A.22})$$

Therefore, by subtracting the blockwise mean before applying $s_j(\cdot)$, we obtain

$$\max_{1 \leq j \leq m_n} |s_j(Y) - s_j(U)| = O_p(m_n^{1/p}(k_n^*\Delta_n)^H + B_{n,R}). \quad (\text{A.23})$$

With the proposed rate of $b = b_n$ in the proof of Theorem 1, all rates on the right-hand sides of (A.19)–(A.23) are $o(1)$. Moreover, since the target variances are bounded away from zero, $s_j(Y) > \sqrt{\Delta_n}$ with probability approaching one. On this event, $\hat{\sigma}_{n,j} = s_j(Y)$ and by $|a/b - 1| = |a^2 - b^2|/(ab + b^2)$, we conclude

$$\max_{1 \leq j \leq m_n} \left| \frac{\hat{\sigma}_{n,j}}{\sigma_{\iota(1,j)\Delta_n}} - 1 \right| = O_p\left(b_n\sqrt{\frac{L_n}{k_n}} + b_n^2\frac{L_n}{k_n} + d_n + m_n^{1/p}(k_n^*\Delta_n)^H + B_{n,R}\right).$$

Let $\mathcal{E}_{n,\sigma}$ be the approximation error on the left-hand side of above display, taking $p > 1/H$ yields $\mathcal{E}_{n,\sigma} = o_p(L_n^{-1})$, which shows uniform consistency of blockwise variance estimators.

Denote

$$\hat{T}_n^* \equiv \max_{1 \leq j \leq m_n} \sup_{t \in \mathcal{T}_{n,j}} \frac{\sqrt{k_{n,j}}|\hat{G}_{n,j} - G_t|}{\hat{\sigma}_{n,j}}, \quad T_n^* = \max_{1 \leq j \leq m_n} |\xi_{n,j}|.$$

Let $\mathcal{E}_{n,G}$ be the approximation error in Theorem 1, and K be the lower bound for true standard deviations stated in Assumption 2(ii). On event $\{\mathcal{E}_{n,\sigma} \leq 1/2\}$, we have

$$\mathcal{E}_{n,T} \equiv |\hat{T}_n^* - T_n^*| \leq \frac{\mathcal{E}_{n,G}}{K} + 2\mathcal{E}_{n,\sigma} \left(T_n^* + \frac{\mathcal{E}_{n,G}}{K} \right).$$

The Gaussian maximal inequality (Lemma 2.2.2 in [van der Vaart and Wellner \(1996\)](#)) gives $T_n^* = O_p(\sqrt{L_n})$. Moreover, since $\mathcal{E}_{n,\sigma} L_n = o_p(1)$ and $\mathcal{E}_{n,G} \sqrt{L_n} = o_p(1)$ by Theorem 1, hence $\mathcal{E}_{n,T} = o_p(L_n^{-1/2})$. It follows Lemma A.1 in [Chernozhukov et al. \(2017\)](#) that, for every deterministic $\delta_n > 0$,

$$\sup_{x \in \mathbb{R}} |\mathbb{P}(\hat{T}_n^* \leq x) - \mathbb{P}(T_n^* \leq x)| \leq \mathbb{P}(\mathcal{E}_{n,T} > \delta_n) + K\delta_n \sqrt{\log(2m_n)}.$$

Let $\delta_n = \epsilon L_n^{-1/2}$. The first term is $o(1)$ and the second term is bounded by $K'\epsilon$. First taking $\Delta_n \rightarrow 0$ then $\epsilon \rightarrow 0$ makes above display vanish. The proof is completed with the fact that $cv_{n,\alpha}$ is the $(1 - \alpha)$ -quantile of T_n^* and

$$\left\{ G_t \in \left[\hat{G}_{n,j} \pm cv_{n,\alpha} \frac{\hat{\sigma}_{n,j}}{\sqrt{k_{n,j}}} \right] \text{ for all } 1 \leq j \leq m_n, t \in \mathcal{T}_{n,j} \right\} = \{ \hat{T}_n^* \leq cv_{n,\alpha} \}.$$

Q.E.D.

Proof of Lemma 2. Similar to the proof of Theorem 1, we show the result under strengthened Assumption 4. We adopt the notation $k_n \equiv \min_{1 \leq j \leq m_n} k_{n,j}$ and $k_n^* \equiv \max_{1 \leq j \leq m_n} k_{n,j}$, further denote

$$\hat{u}_{n,j}(\tau) \equiv \arg \min_{u \in \mathbb{R}} \sum_{i \in \mathcal{I}_{n,j}} \rho_\tau(\mathcal{Y}(\zeta_{i\Delta_n}, \varepsilon_{n,i}) - Q_{i\Delta_n}(\tau) - u).$$

Since τ is fixed in this lemma, we shall suppress the dependence on τ for simplicity. For two vectors a and b , if $\|a - b\|_\infty \leq v$, then $[b]_i - v \leq [a]_i \leq [b]_i + v$, which also applies to every corresponding order statistic. Hence their empirical quantiles differ by at most v . Therefore, pathwise,

$$|\hat{Q}_{n,j} - Q_t - \hat{u}_{n,j}| \leq \sup_{s, s' \in \mathcal{T}_{n,j}} |Q_s - Q_{s'}| + \max_{i \in \mathcal{I}_{n,j}} |R_{n,i}|. \quad (\text{A.24})$$

Assumption 4 implies Q_t is also H -Hölder continuous under the L_p -norm. A similar argument as in (A.8) gives

$$\mathbb{E} \left[\left\{ \max_{1 \leq j \leq m_n} \sup_{s, s' \in \mathcal{T}_{n,j}} \sqrt{k_{n,j}} |Q_s - Q_{s'}| \right\}^p \right] \leq K \Delta_n^{(pH-1)(1-\varrho)-p\varrho/2}.$$

Taking $p > \frac{2(1-\varrho)}{2H(1-\varrho)-\varrho}$ and combining with Assumption 1 gives

$$\max_{1 \leq j \leq m_n} \sup_{t \in \mathcal{T}_{n,j}} \sqrt{k_{n,j}} |\hat{Q}_{n,j} - Q_t - \hat{u}_{n,j}| = o_p(L_n^{-1}).$$

Similarly, let $\bar{f}_{n,j}(\tau) \equiv k_{n,j}^{-1} \sum_{i \in \mathcal{I}_{n,j}} \pi_{n,i}(0, \tau)$, it follows from Assumption 4 that $t \mapsto f_t(Q_t(\tau))$ is also H -Hölder continuous under the L_p -norm. Together with Assumption 5(ii) gives

$$\max_{1 \leq j \leq m_n} |\bar{f}_{n,j} - f_{n,j}| = O_p(d_n + \Delta_n^{(1-\varrho)(H-1/p)}) = o_p(L_n^{-3/2}). \quad (\text{A.25})$$

We shall show the following bounds, the proof is then completed by the triangle inequality.

$$\max_{1 \leq j \leq m_n} |\hat{u}_{n,j}| = O_{a.s.} \left(\sqrt{\frac{L_n}{k_n}} \right), \quad \max_{1 \leq j \leq m_n} \left| \frac{1}{\sqrt{k_{n,j}}} \sum_{i=1}^{k_{n,j}} \frac{U_{i,j}}{f_{n,j}} \right| = O_{a.s.}(\sqrt{L_n}), \quad (\text{A.26})$$

$$\max_{1 \leq j \leq m_n} \left| \sqrt{k_{n,j}} \hat{u}_{n,j} - \frac{1}{\sqrt{k_{n,j}}} \sum_{i=1}^{k_{n,j}} \frac{U_{i,j}}{f_{n,j}} \right| = O_{a.s.}(k_n^{-1/4} L_n^{3/4} + k_n^{-1/2} L_n). \quad (\text{A.27})$$

Retain the empirical CDF and its predictable counterpart,

$$\hat{F}_{n,j}(u) \equiv \frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} \mathbb{1}\{\mathcal{Y}(\zeta_{i\Delta_n}, \varepsilon_{n,i}) - Q_{i\Delta_n}(\tau) \leq u\}, \quad \bar{P}_{n,j}(u) \equiv \frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} P_{n,i}(u).$$

Fix a positive ϖ and let $r_n = K_\varpi \sqrt{L_n/k_n}$ for some positive constant K_ϖ . Note that

$$\{\hat{u}_{n,j} > r_n\} \subset \{\hat{F}_{n,j}(r_n) < \tau\}, \quad \{\hat{u}_{n,j} < -r_n\} \subset \{\hat{F}_{n,j}(-r_n) \geq \tau\}. \quad (\text{A.28})$$

For n large enough, $r_n < u_0/4$, then Assumption 5 implies

$$\bar{P}_{n,j}(r_n) \geq \tau + Kr_n, \quad \bar{P}_{n,j}(-r_n) \leq \tau - Kr_n. \quad (\text{A.29})$$

Conditional on $\mathcal{F}$, for every u , the differences $\mathbb{1}\{\mathcal{Y}(\zeta_{i\Delta_n}, \varepsilon_{n,i}) - Q_{i\Delta_n}(\tau) \leq u\} - P_{n,i}(u)$ are bounded martingale differences with conditional variance at most $1/4$. Combining (A.28) and (A.29), the exponential form of Freedman's inequality (see Theorem 1.6 in Freedman (1975)) gives

$$\begin{aligned} \mathbb{P}(|\hat{u}_{n,j}| > r_n \mid \mathcal{F}) &\leq \mathbb{P}(|\hat{F}_{n,j}(r_n) - \bar{P}_{n,j}(r_n)| \geq Kr_n \mid \mathcal{F}) \\ &\quad + \mathbb{P}(|\hat{F}_{n,j}(-r_n) - \bar{P}_{n,j}(-r_n)| \geq Kr_n \mid \mathcal{F}) \\ &\leq 4 \exp \left\{ -\frac{Kk_{n,j}^2 r_n^2}{2(k_{n,j}/4 + Kk_{n,j}r_n/2)} \right\} \leq 4 \exp \{-K'K_\varpi^2 L_n\}. \end{aligned}$$

Choose K_ϖ such that $K'K_\varpi^2 > 2 + \varpi - \varrho$. A union bound then yields

$$\mathbb{P} \left(\max_{1 \leq j \leq m_n} |\hat{u}_{n,j}| > r_n \mid \mathcal{F} \right) \leq 4m_n \exp \{-K'K_\varpi^2 L_n\} \leq K\Delta_n^{1+\varpi}. \quad (\text{A.30})$$

Summability and the Borel–Cantelli lemma establish the first bound in (A.26). The second follows from the same concentration argument and the score identities $|U_{i,j}| \leq 1$, $\mathbb{E}[U_{i,j} | \mathcal{H}_{n,\iota(i,j)-1}] = 0$, and $\mathbb{E}[U_{i,j}^2 | \mathcal{H}_{n,\iota(i,j)-1}] = \tau(1 - \tau)$.

For (A.27), consider another bounded martingale sequence

$$\begin{aligned} D_{n,i}(u) &\equiv \mathbb{1}\{\mathcal{Y}(\zeta_{i\Delta_n}, \varepsilon_{n,i}) - Q_{i\Delta_n}(\tau) \leq u\} - P_{n,i}(u) \\ &\quad - (\mathbb{1}\{\mathcal{Y}(\zeta_{i\Delta_n}, \varepsilon_{n,i}) - Q_{i\Delta_n}(\tau) \leq 0\} - \tau). \end{aligned}$$

We can verify that $|D_{n,i}(u)| \leq 1$, $\mathbb{E}[D_{n,i}(u) | \mathcal{H}_{n,i-1}] = 0$, and $\mathbb{E}[D_{n,i}(u)^2 | \mathcal{H}_{n,i-1}] \leq K|u|$. Freedman's inequality then gives

$$\mathbb{P} \left(\left| \frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} D_{n,i}(u) \right| > \sqrt{\frac{2Kr_n x}{k_{n,j}}} + \frac{2x}{k_{n,j}} \mid \mathcal{F} \right) \leq 2e^{-x}.$$

Take a deterministic grid $(u_l)_{1 \leq l \leq q_n}$ on $[-r_n, r_n]$ with mesh $K\Delta_n^3$, $q_n \leq K'\Delta_n^{-3}$. Let $x = (5 + \varpi - \varrho)L_n$ above, then a union bound gives

$$\begin{aligned} & \mathbb{P}\left(\max_{1 \leq j \leq m_n} \max_{1 \leq l \leq q_n} \left| \frac{1}{\sqrt{k_{n,j}}} \sum_{i \in \mathcal{I}_{n,j}} D_{n,i}(u_l) \right| > K(k_n^{-1/4} L_n^{3/4} + k_n^{-1/2} L_n) \mid \mathcal{F}\right) \\ & \leq K' m_n q_n \exp\{-(5 + \varpi - \varrho)L_n\} \leq K'' \Delta_n^{1+\varpi}. \end{aligned} \quad (\text{A.31})$$

Note that by definition

$$\frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} D_{n,i}(u) = \widehat{F}_{n,j}(u) - \overline{P}_{n,j}(u) - (\widehat{F}_{n,j}(0) - \tau).$$

By the monotonicity of $\widehat{F}_{n,j}(\cdot)$ and $\overline{P}_{n,j}(\cdot)$, and the fact that $|u_{l+1} - u_l| \leq K\Delta_n^3$, we have for all $u \in [u_l, u_{l+1}]$,

$$\frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} D_{n,i}(u_l) - K\Delta_n^3 \leq \frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} D_{n,i}(u) \leq \frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} D_{n,i}(u_{l+1}) + K\Delta_n^3.$$

Combining with (A.31), it follows Borel–Cantelli lemma that

$$\max_{1 \leq j \leq m_n} \sup_{|u| \leq r_n} \left| \frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} D_{n,i}(u) \right| = O_{a.s.}(k_n^{-3/4} L_n^{3/4} + k_n^{-1} L_n + \Delta_n^3). \quad (\text{A.32})$$

Assumption 5(i) implies the conditional distribution $P_{n,i}$ has no atom on $(-u_0, u_0)$. For any earlier observation, local conditional absolute continuity gives zero probability of a tie in this neighborhood. Taking the countable union over pairs of observations and rows excludes all such local ties at the fixed τ . Together with (A.26) and the smallest-minimizer convention implies eventually for all $1 \leq j \leq m_n$,

$$0 \leq \widehat{F}_{n,j}(\hat{u}_{n,j}) - \tau = \frac{[k_{n,j}\tau]}{k_{n,j}} - \tau < \frac{1}{k_{n,j}}.$$

Moreover, Assumption 5 implies, for $|u| \leq u_0$,

$$|\overline{P}_{n,j}(u) - \tau - \bar{f}_{n,j}u| \leq \frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} \left| \int_0^u (\pi_{n,i}(v) - \pi_{n,i}(0)) dv \right| \leq Ku^2.$$

Since $\widehat{F}_{n,j}(0) - \tau = -k_{n,j}^{-1} \sum_{i=1}^{k_{n,j}} U_{i,j}$, on the event $\{|\hat{u}_{n,j}| \leq r_n\}$, it holds

$$\left| \bar{f}_{n,j} \hat{u}_{n,j} - \frac{1}{k_{n,j}} \sum_{i=1}^{k_{n,j}} U_{i,j} \right| \leq \frac{1}{k_{n,j}} + \sup_{|u| \leq r_n} \left| \frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} D_{n,i}(u) \right| + Kr_n^2. \quad (\text{A.33})$$

Combining (A.30), (A.32), and (A.33) completes the proof of (A.27).

Q.E.D.

Proof of Theorem 2. We adopt the notation from the proof of Lemma 2, and suppress the dependence on τ for simplicity since τ is fixed here. Let $U'_{i,j} \equiv U_{i,j}/\sqrt{k_{n,j}\tau(1-\tau)}$, then $|U'_{i,j}| \leq K_\tau/\sqrt{k_{n,j}}$, $\mathbb{E}[U'_{i,j} \mid \mathcal{H}_{n,\ell(i,j)-1}] = 0$, and $\mathbb{E}[(U'_{i,j})^2 \mid \mathcal{H}_{n,\ell(i,j)-1}] = 1/k_{n,j}$. Following the same idea as the proof of Theorem 1, applying Lemma 1 with $\ell = L_n$ and the conditional Strassen theorem in Cattaneo et al. (2025), we can sequentially construct an extension of $\mathcal{H}$ coupling variables $\xi_{n,j}$ such that $\xi_n \mid \mathcal{F} \sim \mathcal{N}(0, I_{m_n})$ and

$$\mathbb{P}\left(\max_{1 \leq j \leq m_n} \left| \sum_{i=1}^{k_{n,j}} U'_{i,j} - \xi_{n,j} \right| > K k_n^{-1/8} L_n^{5/4}\right) \leq K' m_n \Delta_n = o(1).$$

The proof is completed by setting $Z_{n,j} \equiv \sqrt{\tau(1-\tau)}\xi_{n,j}/f_{n,j}$ and combining with Lemma 2. *Q.E.D.*

Proof of Corollary 2. We adopt the notation from the proof of Theorem 2, and suppress the dependence on τ for simplicity since τ is fixed here. Denote

$$\begin{aligned} \delta_n &\equiv \max_{1 \leq j \leq m_n} \max_{i \in \mathcal{I}_{n,j}} |Q_{i\Delta_n} + R_{n,i} - \hat{Q}_{n,j}|, \\ C_{n,j}(u) &\equiv \frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} \mathbb{1}\{|\mathcal{Y}(\zeta_{i\Delta_n}, \varepsilon_{n,i}) - Q_{i\Delta_n}(\tau)| \leq u\}. \end{aligned}$$

Then we have

$$\frac{C_{n,j}(h_n - \delta_n)}{2h_n} \leq \frac{1}{2h_n k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} \mathbb{1}\{|Y_{n,i} - \hat{Q}_{n,j}| \leq h_n\} \leq \frac{C_{n,j}(h_n + \delta_n)}{2h_n}. \quad (\text{A.34})$$

From (A.24), (A.26), and the triangle inequality, we obtain

$$\begin{aligned} \delta_n &\leq \max_{1 \leq j \leq m_n} |\hat{u}_{n,j}| + 2 \max_{1 \leq j \leq m_n} \sup_{s, s' \in \mathcal{T}_{n,j}} |Q_s - Q_{s'}| + 2 \max_{1 \leq i \leq n} |R_{n,i}| \\ &= O_p(\sqrt{L_n/k_n} + m_n^{1/p} (k_n^* \Delta_n)^H + B_{n,R}) = O_p(\sqrt{L_n/k_n}). \end{aligned}$$

Following the same grid argument in deriving (A.32) to $[-u_0/2, u_0/2]$ and $\mathbb{1}\{\mathcal{Y}(\zeta_{i\Delta_n}, \varepsilon_{n,i}) - Q_{i\Delta_n} \leq u\} - P_{n,i}(u)$ hence the predictable variance is at most $k_{n,j}/4$, it gives

$$\max_{1 \leq j \leq m_n} \sup_{|u| \leq u_0/2} |\hat{F}_{n,j}(u) - \bar{P}_{n,j}(u)| = O_{a.s.}(\sqrt{L_n/k_n}).$$

Combining with Assumption 5, we have uniformly in $1 \leq j \leq m_n$ and $u \in (0, u_0/2]$ that

$$\begin{aligned} C_{n,j}(u) &= \frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} \int_{-u}^u \pi_{n,i}(v) dv + O_{a.s.}(\sqrt{L_n/k_n}) \\ &= 2u \bar{f}_{n,j} + O(u^2) + O_{a.s.}(\sqrt{L_n/k_n}), \end{aligned} \quad (\text{A.35})$$

where the last line uses $\int_{-u}^u |\pi_{n,i}(v) - \pi_{n,i}(0)| dv \leq K u^2$. The rate condition of h_n yields $\delta_n/h_n = o_p(1)$. Therefore, with probability approaching one, $0 < h_n - \delta_n \leq h_n + \delta_n < u_0/2$. Combining

(A.34) and (A.35) gives

$$\begin{aligned} & \max_{1 \leq j \leq m_n} \left| \frac{1}{2h_n k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} \mathbb{1}\{|Y_{n,i} - \hat{Q}_{n,j}| \leq h_n\} - \bar{f}_{n,j} \right| \\ &= O_p \left(\frac{\delta_n}{h_n} + \frac{(h_n + \delta_n)^2}{h_n} + \frac{1}{h_n} \sqrt{\frac{L_n}{k_n}} \right) = O_p \left(h_n + \frac{1}{h_n} \sqrt{\frac{L_n}{k_n}} \right). \end{aligned}$$

Since $\bar{f}_{n,j}$ are bounded away from zero, the floor Δ_n is inactive with probability approaching one. Combining with (A.25), we have

$$\mathcal{E}_{n,f} \equiv \max_{1 \leq j \leq m_n} \left| \frac{\hat{f}_{n,j}}{f_{n,j}} - 1 \right| = O_p \left(h_n + \frac{1}{h_n} \sqrt{\frac{L_n}{k_n}} + d_n \right) = o_p(L_n^{-1}), \quad (\text{A.36})$$

which shows uniform consistency of blockwise density estimators.

Denote

$$\hat{T}_n^* \equiv \max_{1 \leq j \leq m_n} \sup_{t \in \mathcal{T}_{n,j}} \frac{\hat{f}_{n,j} \sqrt{k_{n,j}}}{\sqrt{\tau(1-\tau)}} |\hat{Q}_{n,j} - Q_t|, \quad T_n^* = \max_{1 \leq j \leq m_n} |\xi_{n,j}|.$$

Let $\mathcal{E}_{n,Q}$ be the approximation error in Theorem 2. On event $\{\mathcal{E}_{n,f} \leq 1/2\}$, we have

$$|\hat{T}_n^* - T_n^*| \leq K \mathcal{E}_{n,Q} + 2 \mathcal{E}_{n,f} (T_n^* + K \mathcal{E}_{n,Q}).$$

The rest of the proof is the same as in Corollary 1.

Q.E.D.

Proof of Theorem 3. We use the same localization reduction and notation as in the proof of Lemma 2, restoring the dependence on τ . All density bounds below hold uniformly over $\tau \in [\underline{\tau}, \bar{\tau}]$ and a common neighborhood of the corresponding quantiles. Under Assumption 6, we have

$$P_{n,i}(u, \tau) = F(\zeta_{i\Delta_n}, Q_{i\Delta_n}(\tau) + u), \quad \bar{f}_{n,j}(\tau) = \frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} f_{i\Delta_n}(Q_{i\Delta_n}(\tau)),$$

so the predictable density equals the state-based conditional density and density stabilization is exact. The density lower bound implies $|Q_t(\tau) - Q_t(\tau')| \leq K|\tau - \tau'|$ for $\tau, \tau' \in [\underline{\tau}, \bar{\tau}]$.

We first show a uniform Bahadur representation akin to Lemma 2 is valid for $\tau \in [\underline{\tau}, \bar{\tau}]$. Extend $\hat{F}_{n,j}$, $\bar{P}_{n,j}$, and $D_{n,i}$ from the proof of Lemma 2 by retaining their τ argument. Thus

$$\frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} D_{n,i}(u, \tau) = \hat{F}_{n,j}(u, \tau) - \bar{P}_{n,j}(u, \tau) - (\hat{F}_{n,j}(0, \tau) - \tau).$$

Set $r_n = K\sqrt{L_n/k_n}$, with K sufficiently large. As in deriving (A.30) and (A.32), use deterministic grids of mesh at most $K'\Delta_n^3$, now on both $[\underline{\tau}, \bar{\tau}]$ and $[-2r_n, 2r_n]$. There are $O(\Delta_n^{-6})$ grid pairs. Let $s \leq \tau \leq t$ and $v \leq u \leq w$ be adjacent grid points. Monotonicity of $\hat{F}_{n,j}(\cdot, \cdot)$ gives

$$\hat{F}_{n,j}(v, s) - \hat{F}_{n,j}(0, t) \leq \hat{F}_{n,j}(u, \tau) - \hat{F}_{n,j}(0, \tau) \leq \hat{F}_{n,j}(w, t) - \hat{F}_{n,j}(0, s),$$

where the predictable means of these brackets differ from $\bar{P}_{n,j}(u, \tau) - \tau$ by at most $K\Delta_n^3$. Each bracket is an average of differences of nested indicators and its centered summands are uniformly

bounded and have conditional variances bounded by $K(r_n + \Delta_n^3)$. Thus the same Freedman calculation applies to the brackets. Taking its parameter to be KL_n with $K > 8 - \varrho$ bounds the union probability by $K'\Delta_n^{K-7+\varrho}$, which is summable. The quantile-localization union is also summable after increasing the constant in r_n . Borel–Cantelli lemma therefore gives

$$\max_{1 \leq j \leq m_n} \sup_{\tau \in [\underline{\tau}, \bar{\tau}]} |\hat{u}_{n,j}(\tau)| \leq r_n, \quad \text{eventually almost surely.} \quad (\text{A.37})$$

The quantile-localization argument also extends to all τ by bracketing each quantile level with adjacent grid points. Since the predictable separation is at least Kr_n , whereas the quantile-grid error is $O(\Delta_n^3)$, we have

$$\begin{aligned} & \max_{1 \leq j \leq m_n} \sup_{\tau \in [\underline{\tau}, \bar{\tau}]} \sup_{|u| \leq r_n} \left| \frac{1}{k_{n,j}} \sum_{i \in \mathcal{I}_{n,j}} D_{n,i}(u, \tau) \right| \\ &= O_{a.s.}(k_n^{-3/4} L_n^{3/4} + k_n^{-1} L_n + \Delta_n^3). \end{aligned} \quad (\text{A.38})$$

For inversion uniformly in τ , note that $\hat{F}_{n,j}(\hat{u}_{n,j}(\tau) - , \tau) \leq \tau \leq \hat{F}_{n,j}(\hat{u}_{n,j}(\tau), \tau)$ and the predictable CDF is continuous, which avoids an uncountable no-ties argument. By the same argument as in (A.33), using $|\bar{P}_{n,j}(u, \tau) - \tau - \bar{f}_{n,j}(\tau)u| \leq Ku^2$, together with (A.38) we obtain

$$\begin{aligned} & \max_{1 \leq j \leq m_n} \sup_{\tau \in [\underline{\tau}, \bar{\tau}]} \left| \sqrt{k_{n,j}} \hat{u}_{n,j}(\tau) - \frac{1}{\bar{f}_{n,j}(\tau) \sqrt{k_{n,j}}} \sum_{i=1}^{k_{n,j}} U_{i,j}(\tau) \right| \\ &= O_{a.s.}(k_n^{-1/4} L_n^{3/4} + k_n^{-1/2} L_n), \end{aligned} \quad (\text{A.39})$$

where the last line uses $k_n \Delta_n^3 = o(1)$. Together with (A.37), this also gives

$$\max_{1 \leq j \leq m_n} \sup_{\tau \in [\underline{\tau}, \bar{\tau}]} \left| \frac{1}{\sqrt{k_{n,j}}} \sum_{i=1}^{k_{n,j}} U_{i,j}(\tau) \right| = O_{a.s.}(\sqrt{L_n}). \quad (\text{A.40})$$

We then construct a joint Brownian-bridge coupling for $k_{n,j}^{-1/2} \sum_{i=1}^{k_{n,j}} U_{i,j}(\tau)$. Define $V_{n,i} = F(\zeta_{i\Delta_n}, \mathcal{Y}(\zeta_{i\Delta_n}, \varepsilon_{n,i}))$. Conditional independence and continuity of the marginal CDFs imply that $(V_{n,i})_{1 \leq i \leq n}$ are conditionally i.i.d. uniform given $\mathcal{F}$. Moreover, simultaneously for all $\tau \in [\underline{\tau}, \bar{\tau}]$,

$$U_{i,j}(\tau) = \tau - \mathbb{1}\{V_{n,\iota(i,j)} \leq \tau\}.$$

Applying the empirical-process coupling of Komlós et al. (1975) in each block yields standard Brownian bridges $B_{n,j}^0$ such that, for every $x > 0$,

$$\mathbb{P} \left(\sup_{v \in [0,1]} \left| \frac{1}{\sqrt{k_{n,j}}} \sum_{i=1}^{k_{n,j}} (v - \mathbb{1}\{V_{n,\iota(i,j)} \leq v\}) - B_{n,j}^0(v) \right| > \frac{K \log k_{n,j} + x}{\sqrt{k_{n,j}}} \mid \mathcal{F} \right) \leq K' e^{-Kx}.$$

Realize a fixed Komlós et al. (1975) joint law through the conditional kernel for the bridge given the observed block ranks, using independent auxiliary randomization across blocks. This

preserves the original observations. The product law given $\mathcal{F}$ then makes the resulting bridges jointly conditionally independent. Taking $x = ML_n$ where $M > 0$, a union bound gives

$$\mathbb{P}\left(\max_{1 \leq j \leq m_n} \sup_{\tau \in [\underline{\tau}, \bar{\tau}]} \left| \frac{1}{\sqrt{k_{n,j}}} \sum_{i=1}^{k_{n,j}} U_{i,j}(\tau) - B_{n,j}^0(\tau) \right| > \frac{K_M L_n}{\sqrt{k_n}} \mid \mathcal{F}\right) \leq K' m_n e^{-K M L_n}.$$

Choose M sufficiently large such that the right-hand side is summable, Borel–Cantelli lemma therefore yields

$$\max_{1 \leq j \leq m_n} \sup_{\tau \in [\underline{\tau}, \bar{\tau}]} \left| \frac{1}{\sqrt{k_{n,j}}} \sum_{i=1}^{k_{n,j}} U_{i,j}(\tau) - B_{n,j}^0(\tau) \right| = O_{a.s.}\left(\frac{L_n}{\sqrt{k_n}}\right). \quad (\text{A.41})$$

Finally, repeating the derivation of (A.24) and (A.25), now uniformly in τ , gives

$$\begin{aligned} & \max_{1 \leq j \leq m_n} \sup_{\tau \in [\underline{\tau}, \bar{\tau}]} \sup_{t \in \mathcal{T}_{n,j}} \sqrt{k_{n,j}} |\hat{Q}_{n,j}(\tau) - Q_t(\tau) - \hat{u}_{n,j}(\tau)| \\ &= O_p\left(\sqrt{k_n} m_n^{1/p} (k_n^* \Delta_n)^H + \sqrt{k_n} B_{n,R}\right), \\ & \max_{1 \leq j \leq m_n} \sup_{\tau \in [\underline{\tau}, \bar{\tau}]} |\bar{f}_{n,j}(\tau) - f_{n,j}(\tau)| = O_p\left(m_n^{1/p} (k_n^* \Delta_n)^H\right). \end{aligned} \quad (\text{A.42})$$

The proof is completed by taking $p > \frac{2(1-\varrho)}{2H(1-\varrho)-\varrho}$ and combining (A.39)–(A.42). Q.E.D.

Proof of Corollary 3. Work after the same localization reduction and use the coupling in Theorem 3. Since $\text{cv}_{n,\alpha} = O(\sqrt{L_n})$ and k_n grows polynomially with Δ_n^{-1} , the shifted quantile levels lie in $[\underline{\tau}, \bar{\tau}]$ for all sufficiently large Δ_n^{-1} .

The usual Brownian increment bound and a union bound over j give, for every fixed K ,

$$\max_{1 \leq j \leq m_n} \sup_{|v-\tau| \leq K\sqrt{L_n/k_n}} |B_{n,j}^0(v) - B_{n,j}^0(\tau)| = O_p(k_n^{-1/4} L_n^{3/4}). \quad (\text{A.43})$$

Uniform local inverse-CDF regularity gives

$$Q_t(v) - Q_t(\tau) = \frac{v - \tau}{f_t(Q_t(\tau))} + O((v - \tau)^2).$$

The density freezing in (A.42) permits replacing the density at t by its block-anchor value, with polynomially negligible error. Combining with the full process coupling and (A.43) yields

$$\begin{aligned} & f_{n,j}(\tau) \sqrt{k_{n,j}} \left\{ \hat{Q}_{n,j} \left(\tau \pm \text{cv}_{n,\alpha} \sqrt{\frac{\tau(1-\tau)}{k_{n,j}}} \right) - Q_t(\tau) \right\} \\ &= B_{n,j}^0(\tau) \pm \text{cv}_{n,\alpha} \sqrt{\tau(1-\tau)} + o_p(L_n^{-1/2}) + O_p(k_n^{-1/4} L_n^{3/4}) + O_p(L_n k_n^{-1/2}). \end{aligned}$$

The additional errors above are all $o_p(L_n^{-1/2})$.

Note that $B_{n,j}^0(\tau)/\sqrt{\tau(1-\tau)}$ is $\mathcal{F}$ -conditional standard normal and independent across j , hence we can take $\xi_{n,j} = B_{n,j}^0(\tau)/\sqrt{\tau(1-\tau)}$. For any fixed $\epsilon > 0$, the simultaneous coverage event is included, with probability approaching one, between

$$\left\{ \max_{1 \leq j \leq m_n} |\xi_{n,j}| \leq \text{cv}_{n,\alpha} - \epsilon L_n^{-1/2} \right\} \quad \text{and} \quad \left\{ \max_{1 \leq j \leq m_n} |\xi_{n,j}| \leq \text{cv}_{n,\alpha} + \epsilon L_n^{-1/2} \right\}.$$

By the same Gaussian anti-concentration argument as in Corollary 1, the difference between these two probabilities is at most $K\epsilon$. First let $\Delta_n \rightarrow 0$ and then $\epsilon \rightarrow 0$. The definition of $\text{cv}_{n,\alpha}$ gives the claimed coverage. *Q.E.D.*

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